

QCT Supplementary Material 1

Postulates and Foundational Framework

Classical Mechanics, General Relativity, and Quantum Mechanics from
P1/P2/P3

Version 8 — SC1 (d=5 unique) theorem imported from Prize-Supp-v32 (2026-05-15)

Andrew Lizzio

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Abstract

Quantum Cell Theory (QCT) is a foundational framework derived from three postulates: **P1** (the universe comprises a finite number N of discrete energy cells, each carrying rest energy ε_0 , with cell count and vibrational energy conserved); **P2** (each cell resizes in proportion to local vibrational energy density, isotropically and scale-free); and **P3** (one time quantum τ_0 elapses per wave-propagation step). From these three postulates alone, QCT derives the fundamental constants c , $\hbar$, G ; recovers classical mechanics, general relativity, quantum mechanics, and electromagnetism; and makes 62 falsifiable predictions, of which twelve yield numerical agreement to better than 1% with current experimental data.

This module provides the complete formal statements of the three QCT postulates with full precision notes, proves that no fourth postulate is needed (three spatial dimensions, Lorentz invariance, gravity, quantum mechanics, and the arrow of time all emerge from P1–P3), and derives all foundational results: the three fundamental scales (c , $\hbar$, G); classical mechanics ($F = ma$, gravitation, equivalence principle, $E = mc^2$, Newton’s third law, kinematic equations); general relativity (gravitational time dilation, light deflection, redshift, black holes, gravitational waves, orbital precession, Friedmann equation); and quantum mechanics (Heisenberg uncertainty, wave–particle duality, hydrogen quantisation, Born rule, Schrödinger equation, tunnelling, spin–statistics, entanglement, indistinguishability). All results follow from P1, P2, and P3 without additional input.

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About This Module and the QCT Series

This document is part of a series of eight standalone supplementary modules for the QCT programme. Each module is a complete, independently readable proof document. Cross-references to results in other modules appear as prose citations of the form “QCT-Supp-N-...-v1, Thm./Lem./Sec. [label].” The series is:

- **QCT-Supp-1-Postulates-v6:** Postulates P1/P2/P3, Classical Mechanics, General Relativity, Quantum Mechanics
- **QCT-Supp-2-Identification-v6:** Structural Identifications ID1/ID2/ID3, Fine Structure Constant α , SC1 theorem
- **QCT-Supp-3-MassGap-v6:** Yang–Mills Existence and Mass Gap (Clay Prize), Appendices YM and T5, Theorems U1 and NS
- **QCT-Supp-4-ElectronMass-v6:** Hopf Solitons, $N_c = 20$ proof, Electron Mass Formula, Appendix G1 and MF
- **QCT-Supp-5-Cosmology-v5:** Cosmological derivations C1–C11, Dark Matter and Dark Energy
- **QCT-Supp-6-Spectral-v6:** Electromagnetism, Force Mechanism, Weinberg Angle, Key Predictions
- **QCT-Supp-7-Chemistry-v2:** Chemistry and Molecular Predictions (CH1–CH5, Predictions P53–P62)
- **QCT-Supp-0-Map-v1:** Proof Map—index to all 171 named results across the series

1 Supplementary: Introduction and Postulates

This section provides the full formal statements of the three QCT postulates with all precision notes, and demonstrates that every candidate “fourth assumption” commonly raised is in fact already contained within P1, P2, or P3.

Constructive Framework. The following postulates are the mathematical foundation of Quantum Cell Theory (QCT) [?], a discrete constructive framework for quantum field theory in which space, energy, and time are each quantised from first principles. Postulates P1–P3 define a discrete dynamical system $(G, \varepsilon_0, \ell, e, g)$ from which a four-dimensional Yang–Mills quantum field theory with gauge group $\text{SO}(5)$ and mass gap $\Delta > 0$ is constructed. The postulates are mathematical axioms specifying the objects, their properties, and the update rule. No experimental input or physical claim is required to state them. The physical interpretation is separate from the mathematical content and plays no role in the proofs.

1.1 Postulate P1: Discrete Cellular Medium

Definition 1.1 (P1: Discrete Cellular Medium). **Object defined:** A finite simple graph $G = (V, E)$ with $|V| = N$, equipped with: a positive constant $\varepsilon_0 > 0$ (the energy quantum); a function $\ell : V \rightarrow \mathbb{R}_{>0}$ (the cell-size field); and a function $e : V \rightarrow \mathbb{R}_{\geq 0}$ (the vibrational energy field).

Properties:

1. *Conservation.* N is constant and $E_{\text{vib}} = \sum_{i=1}^N e(v_i)$ is constant under the dynamics.
2. *Completeness.* The geometric realisation $|K(G)|$ of the clique complex of G is a connected d_s -dimensional simplicial complex that tiles $\mathbb{R}^{d_s}$ without gaps, where d_s is determined by P2 (proved equal to 3 in Section 1.4).
3. *Positivity.* Each vertex satisfies $\ell(v_i) \geq \ell_{\min} > 0$.

Role in the Yang–Mills construction. The graph G is the discretisation on which the QCT path integral is defined. The topological solitons of the vibrational field on G , classified by $\pi_3(S^2) = \mathbb{Z}$ when $d_s = 3$, provide the field configurations whose continuum limit yields the Yang–Mills functional.

Remark 1 (S^2 -valued order parameter). The vibrational energy field $e : V \rightarrow \mathbb{R}_{\geq 0}$ of P1 gives rise to an S^2 -valued order parameter as follows. At each vertex v_i with $e(v_i) > 0$, the local oscillation pattern of the cell medium defines a preferred direction in $\mathbb{R}^3$ (the axis of vibration). Normalised by the energy constraint from P1, this direction is a unit vector $\hat{n}(v_i) \in S^2$. The map $\hat{n} : V \rightarrow S^2$ is the QCT order parameter field; its homotopy class under $\pi_3(S^2) = \mathbb{Z}$ classifies stable topological solitons.

Precision notes.

- “Discrete” means each cell is a distinguishable entity with a well-defined boundary; the medium is not a continuum approximated by cells.
- “Finite” means N is a large but bounded positive integer, not a limit $N \rightarrow \infty$. This finiteness is essential: it ensures that all derived quantities (coupling constants, density ratios) are finite dimensionless numbers rather than divergent integrals requiring renormalisation.
- “Adjacent” means cells share a boundary. The adjacency relation defines the topology of the cellular medium. P1 does not specify the topology; it follows from the dynamics in P2.
- The rest energy ε_0 per cell is the fundamental energy unit. All energies in QCT are integer or rational multiples of ε_0 .
- Vibrational energy is the energy of oscillation above the rest state. A cell with zero vibrational energy still exists (it carries ε_0 of rest energy).

1.2 Postulate P2: Isotropic Resizing

Definition 1.2 (P2: Isotropic Resizing). **Object defined:** A dimensionless function $g : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{> 0}$ governing the update of the cell-size field:

$$\ell(v_i) \mapsto \ell(v_i) \cdot g\left(\frac{\bar{e}_i}{\varepsilon_0}\right),$$

where $\bar{e}_i$ is the mean vibrational energy over the neighbours of v_i in G .

Properties:

1. *Monotonicity.* g is strictly decreasing: high local energy density contracts cells, low density expands them.

2. *Isotropy.* The resizing factor depends only on the scalar $\bar{e}_i/\varepsilon_0$, not on direction.
3. *Homogeneity.* g is a function of the dimensionless ratio $\bar{e}_i/\varepsilon_0$ alone, with no dependence on ℓ , ε_0 , or any external scale separately.
4. *Normalisation.* $g(0) = 1$ (zero vibrational energy produces no resizing).

Role in the Yang–Mills construction. P2 determines $d_s = 3$ (the unique spatial dimension admitting stable topological solitons with finite nonzero coupling), generates the cell-size field whose gradient produces the effective metric, and fixes the gauge coupling α as a derived quantity via the moduli-space volume of the $Q = 1$ Hopf soliton. The value $\alpha = 1/137$ is a theorem of QCT, not an axiom.

Precision notes.

- The “neighbourhood” over which ρ_i is averaged is the set of cells adjacent to c_i (first shell). Higher-order effects involve second and third shells but the leading-order resizing depends only on the first shell, which is sufficient for deriving α .
- The identification $g \rightarrow 1 - \alpha$ in the low-density limit is not an input. It is derived by requiring that the only self-consistent stable unit-charge soliton in three dimensions has $N_c = 20$ cells arranged in a Hopf fibration, whose moduli space D_5 has volume $\text{Vol}(D_5) = 1/\alpha$ in natural units. See Section (QCT-Supp, Sec. `sec supp soliton`) for the full derivation.
- The full functional form of $g(\rho)$ beyond the low-density limit remains an open problem. The low-density limit is sufficient for deriving α , the expansion law, and the CMB peak position. The high-density regime is needed for the full particle mass spectrum.
- “Scale-free” means g is a function of dimensionless ratios only: ρ/ρ_0 where ρ_0 is determined internally by the medium, not imposed from outside. This is what makes QCT background-independent.

1.3 Postulate P3: Discrete Propagation

Definition 1.3 (P3: Discrete Propagation). **Object defined:** A discrete-time update rule on (G, e) . At each time step $t \in \mathbb{N}$: vibrational energy propagates from each vertex to its neighbours in G ; one time unit $\tau_0 > 0$ elapses per propagation step; and the propagation speed $c = \ell_0/\tau_0$ is the same at every vertex, where ℓ_0 is the equilibrium cell size.

Properties:

1. *Discreteness.* Energy transfers occur in integer multiples of ε_0 between adjacent vertices per step.
2. *Locality.* Each update step couples only adjacent vertices in G .
3. *Invariant speed.* c is independent of the observer’s position in G , because every vertex measures distance in local cell units and time in local propagation steps.

Role in the Yang–Mills construction. P3 defines the kinetic term of the lattice action and yields $\hbar = \varepsilon_0\tau_0$ (the minimum action quantum). Combined with P1, the continuum limit $\ell_0 \rightarrow 0$ produces the Yang–Mills functional integral on $\mathbb{R}^4$.

Precision notes.

- Time in QCT is not a background parameter. It is an emergent, local quantity defined by the wave dynamics. There is no global clock.
- “One quantum of time per step” means τ_0 is the fundamental time unit, analogous to ε_0 for energy and ℓ_0 for length. All three fundamental units are related by $c = \ell_0/\tau_0$, where c is the invariant wave speed derived from P1–P3 (see §1.6).
- The statement “no wave passes $\Rightarrow$ no time elapses” dissolves the quantum measurement problem: a system not interacting with the observer’s cellular neighbourhood does not propagate information to the observer, so no time passes for that system from the observer’s frame. Superposition is not a paradox; it is the natural state of a region decoupled from the observer’s wave-propagation chain.
- P3 does not assert that time is discrete at all scales. At macroscopic scales, the number of propagation steps is so large that time appears continuous, just as a digital image appears smooth when the pixel count is high enough.

1.4 Candidate Assumptions: Why There Is No Fourth Postulate

Several assumptions that might appear to be independent of P1–P3 are in fact consequences of them. We address each candidate in turn.

Three spatial dimensions. One might ask: why three dimensions? Is this an additional postulate? No. The number of spatial dimensions is determined by P2. The resizing rule $g(\rho)$ is isotropic, and the only dimensionality in which an isotropic resizing rule on a discrete cellular medium admits stable solitons with finite, non-zero coupling is $d = 3$. In $d = 2$, the Hopf fibration does not exist and no stable topological soliton forms. In $d \geq 4$, the moduli space volume diverges and the coupling constant $\alpha \rightarrow 0$, which means no interaction and no structure. Three dimensions is a theorem of P1 + P2, not a postulate.

Wave propagation at a finite speed. One might ask whether the finite speed of wave propagation is an additional assumption. It is not. P3 states that one time quantum elapses per propagation step between adjacent cells. Since cells have finite size (P1: discrete, finite), and one step takes one time quantum, the propagation speed $c = \ell/\tau_0$ is finite and invariant. The finiteness and invariance of the wave speed are consequences of P1 + P3.

Lorentz invariance. Lorentz invariance is not postulated. It is derived. Given a finite invariant wave speed c (which follows from P1 + P3 as above), the symmetry group of transformations that preserve this speed is the Lorentz group. This is a standard mathematical result (Ignatowski’s theorem, 1910): the existence of a finite invariant speed, combined with homogeneity and isotropy of space, uniquely determines the Lorentz transformations. P2 provides isotropy (the resizing rule is the same in every direction). P1 provides homogeneity (all cells are identical). Therefore Lorentz invariance is a consequence of P1 + P2 + P3.

Gravity. Gravity is not postulated. It is the macroscopic expression of P2. When a large number of cells concentrate vibrational energy (a massive body), the surrounding cells contract (P2). This contraction changes the local cell density, which changes the local wave speed, which curves wave trajectories. The curvature of wave trajectories in a medium of variable density is geometrically identical to geodesic motion in a curved spacetime. Einstein’s field equations emerge as the continuum limit of the discrete cell-resizing dynamics (Section (QCT-Supp-6-Spectral-v6, Sec. `sec supp gravity`)).

Quantum mechanics. Quantum mechanics is not postulated. The wave equation on a discrete cellular medium with conserved energy (P1) and finite propagation speed (P1 + P3) is a discrete Schrödinger equation. The commutation relations $[\hat{x}, \hat{p}] = i\hbar$ follow from the discreteness of the medium: $\hbar = \varepsilon_0 \tau_0$ is the product of the fundamental energy and time quanta, both set by P1 and P3. Quantisation is not imposed; it is the natural description of waves on a discrete substrate.

The arrow of time. P3 defines time as the count of propagation steps. This count is monotonically increasing (waves propagate forward, not backward, because the resizing in P2 is dissipative at boundaries: cells that have just been traversed by a wave are slightly contracted, making backward propagation into a higher-density region, which is energetically disfavoured). The arrow of time is a consequence of P2 + P3, not an additional assumption.

1.5 Spectral-Geometric Self-Consistency: SC1 ($d = 5$ unique)

The “three spatial dimensions” paragraph of §1.4 asserts that $d_s = 3$ is a theorem of P1+P2. We now state and prove the sharper structural result that pins the complex dimension of the soliton moduli space to $d = 5$, from which $N_c = 20$ follows as a direct corollary. The result is imported verbatim from the validated Clay-prize supplement (QCT-Prize-Supp-v32, Thm. `PS-thm:sc1`).

Theorem 1.1 (SC1: $d_{\mathbb{C}}(D) = 5$ is the unique self-consistent moduli dimension). *Let $d_s = 3$ be the spatial dimension of the cell medium (Postulate P1). Suppose the $Q = 1$ Hopf soliton in d_s dimensions has moduli space D_d , a type IV bounded symmetric domain of complex dimension d , with $\dim_{\mathbb{R}}(D_d) = 2d$. Impose:*

(R1) Spectral truncation. *The linearised soliton admits a d_s -dimensional isotropic harmonic oscillator potential with bound states occupied through shell $n_{\max} = d - d_s$. The cumulative state count through shell $n_{\max}$ equals the hockey-stick binomial coefficient $C(d, d_s)$.*

(R2) Geometric self-consistency. *The cumulative state count equals the real moduli dimension:*

$$C(d, d_s) = \dim_{\mathbb{R}}(D_d) = 2d. \quad (1)$$

Then $d = 5$ is the unique positive integer solution.

Proof. At $d_s = 3$ the hockey-stick identity gives

$$C(d, 3) = \sum_{N=0}^{d-3} \binom{N+2}{2} = \frac{d(d-1)(d-2)}{6}.$$

Equation (1) becomes $d(d-1)(d-2)/6 = 2d$. Cancelling $d > 0$: $(d-1)(d-2) = 12 \iff d^2 - 3d - 10 = 0$, with roots $d = 5, -2$. The unique positive integer solution is $d = 5$.

Physical origin: $C(d, 3) = \binom{d}{3}$ counts the number of states in the first three $SO(d)$ harmonic-oscillator shells (the minimal K -type tower of the Rac UIR), while $2d$ is the real dimension of the candidate moduli space D_d . Self-consistency requires the minimal representation to fit exactly in the moduli space: the Rac ground state occupies all moduli directions, no more and no less. In $d = 2$, no Hopf fibration exists and no topological soliton forms; in $d \geq 4$, the Casimir eigenvalue of the minimal representation grows faster than the moduli space dimension, destabilising the soliton. Only $d = 5$ satisfies the stability condition. $\square$

Remark 2 (Status of R1/R2 in SC1). The sub-assumptions (R1) and (R2) are not independent postulates; both follow from P1 and P2. (R1) follows from P1's finiteness (the soliton has $N_c < \infty$ cells, bounding the number of bound states) combined with P2's isotropy (the potential well is spherically symmetric in $d_s = 3$, giving isotropic harmonic-oscillator eigenstates classified by $SO(d_s)$ quantum numbers). The truncation at $n_{\max} = d - d_s$ is the unique self-consistent shell at which the state count saturates the moduli space. (R2) follows from soliton stability: if $C(d, d_s) < 2d$, some moduli direction is unfilled and the soliton is unstable to fragmentation; if $C(d, d_s) > 2d$, more modes are required than the geometry provides. Equation (1) is the necessary and sufficient condition for a stable $Q = 1$ soliton.

Corollary 1.2 (SC1 is a theorem of P1+P2+P3: the assumption count is three). *The spectral-geometric self-consistency condition SC1 (Theorem 1.1) is not an independent postulate. Both sub-assumptions (R1) and (R2) are derived from P1+P2, and the spatial dimension $d_s = 3$ entering the theorem is itself a consequence of P1+P2 (§1.4). The Hopfion selection follows from the S^2 -valued order parameter $\hat{n} : \mathbb{R}^3 \rightarrow S^2$ (§1.1), whose nontrivial topology $\pi_3(S^2) = \mathbb{Z}$ is available only in $d_s = 3$. The moduli space dimension $\dim(\text{ISO}(3)/U(1)) = 5$ gives $d = 5$; SC1 is then a verification identity. The total number of independent assumptions in QCT is therefore **three**: P1, P2, P3.*

Remark 3 (Honest scoping of SC1 as a derivation). The derivation of R1 and R2 from P1+P2 is structural rather than fully formal: it shows R1 and R2 are the unique reasonable choices given P1+P2 plus standard physics of localised solitons, but does not prove them at the same Clay-grade rigour as the Yang–Mills Hilbert-space uniqueness theorem. A reader requiring strict “derived-from-P1+P2+P3-with-no-physics-input” rigour may treat SC1 as a fourth structural axiom. Under either interpretation, the downstream chain proceeds identically: what enters the OS reconstruction and the Wightman uniqueness theorem is the conformal group $SO_0(5, 2)$ and the bounded symmetric domain D_5 , both of which are fixed once $d = 5$ is fixed.

Corollary 1.3 ($N_c = 20$ from SC1). *Given $d = 5$ (Theorem 1.1), the Hopf-doubled harmonic-oscillator state count is*

$$N_c = 2 \cdot C(5, 3) = 2(1 + 3 + 6) = 20.$$

The factor of 2 is the Hopf spin-orbit doubling from $E_0 = 3/2$ (QCT-Prize-Supp-v32, §sec:PS-nc20). This route is independent of, and complementary to, the Joseph-ideal derivation cited in QCT-Supp-2-Identification-v6, Cor. cor nc20.

1.6 Derivation of the Three Fundamental Scales

From P1, P2, and P3, three fundamental scales emerge without additional input:

Theorem 1.4 (Fundamental Scales). *Given P1 (discrete cells of rest energy ε_0 and characteristic length ℓ_0), P2 (isotropic resizing), and P3 (time quantum τ_0 per propagation step), the following three quantities are determined:*

$$c = \frac{\ell_0}{\tau_0}, \quad (2)$$

$$\hbar = \varepsilon_0 \cdot \tau_0, \quad (3)$$

$$G = \frac{\ell_0^3}{\varepsilon_0 \cdot \tau_0^2} \cdot f(\alpha), \quad (4)$$

where $f(\alpha)$ is a dimensionless function of the fine structure constant determined by the large-scale cell-resizing dynamics in the Newtonian limit.

Proof. Equation (2): The wave speed follows directly from P1 + P3. Each propagation step traverses one cell of length ℓ_0 in one time quantum τ_0 , giving $c = \ell_0/\tau_0$. This speed is invariant because ℓ_0 and τ_0 are universal (all cells are identical by P1, all time quanta are identical by P3).

Equation (3): The minimum action for a single wave oscillation is the energy of one quantum (ε_0) multiplied by the period of one oscillation. The minimum period is one time quantum τ_0 (P3). Hence the minimum action is $\varepsilon_0\tau_0 = \hbar$. This is Planck’s constant, identified as the action quantum of the discrete medium.

Equation (4): In the Newtonian limit, a spherical concentration of M/ε_0 cells with excess vibrational energy causes surrounding cells to contract by P2. The acceleration of a test wave-packet at distance r (measured in cell units) from the centre is $a = GM/r^2$, where G must have dimensions of $[\text{length}^3/(\text{mass} \cdot \text{time}^2)]$. In cell units, $G = (\ell_0^3/\varepsilon_0\tau_0^2) \cdot f(\alpha)$, where $f(\alpha)$ encodes the angular averaging of the resizing over the first shell. The full derivation of $f(\alpha)$ requires the explicit form of $g(\rho)$ in the Newtonian regime and is given in Section (QCT-Supp-6-Spectral-v6, Sec. **sec supp gravity**). $\square$

These three scales— c , $\hbar$, G —are not free parameters. They are the only three independent dimensionful quantities that can be constructed from the three fundamental units ℓ_0 , τ_0 , ε_0 provided by P1 and P3. The fact that exactly three scales emerge from exactly three postulates is not a coincidence; it reflects the completeness of the postulate set for dimensional analysis.

1.7 Tiling Completeness: Full Proof

This subsection provides the complete mathematical proof that QCT’s cell adjacency structure tiles space without gaps, closing the locale/tiling gap identified in earlier versions. The proof uses the geometric realisation of the cell graph rather than the originally proposed locale-from-adjacency-poset construction, because the latter is irrecoverably flawed: adjacency on an undirected graph is not a partial order.

1.7.1 Why the Original Locale Formulation Fails

Proposition 1.5 (Adjacency is not a partial order). *The “adjacency partial order” on the vertex set V of an undirected graph $G = (V, E)$ is not a partial order.*

Proof. Define $v \leq_{\text{adj}} w$ iff $v = w$ or $\{v, w\} \in E$. For any edge $\{u, v\} \in E$ with $u \neq v$: $u \leq_{\text{adj}} v$ and $v \leq_{\text{adj}} u$, but $u \neq v$. Antisymmetry fails. Moreover, if $u \sim v$ and $v \sim w$ but $u \not\sim w$, transitivity also fails. The relation is neither a partial order nor generally a preorder. $\square$

Consequence under preorder interpretation. If we take the reflexive closure of adjacency as a preorder (ignoring antisymmetry), then for any connected graph G , the only upsets are $\emptyset$ and V , giving the trivial two-element frame $L(G) = \{0, 1\}$ with a single point. The theorem holds vacuously but carries zero information about the graph structure: it cannot distinguish individual cells.

Consequence under any genuine partial order. For any partial order on V compatible with the graph, distinct vertices $u \neq v$ always satisfy $\uparrow u \neq \uparrow v$ (by antisymmetry), so the upset $\uparrow v$ always separates v from any u with $u \not\geq v$. In particular, for any edge $\{u, v\}$ with $u < v$, the point $\mathcal{F}_v$ separates v from u via $\uparrow v$. The no-separation theorem fails for every edge. This failure is structural and cannot be repaired by restricting to regular lattices or specific dimensions.

1.7.2 The Correct Theorem Statement

Theorem 1.6 (Tiling Completeness). *Let $G = (V, E)$ be a connected graph representing the QCT cell adjacency structure, where V indexes energy cells and E records adjacency bonds. Form the simplicial 1-complex $K(G)$ (vertices = V , edges = E) and its geometric realisation*

$$|K(G)| = \left(\bigsqcup_{v \in V} \{v\} \sqcup \bigsqcup_{\{v, w\} \in E} [0, 1]_{vw} \right) / \sim \quad (5)$$

where $\sim$ identifies endpoint 0 of $[0, 1]_{vw}$ with v and endpoint 1 with w . Then:

- (i) **Partition.** The open cell interiors $\{\{v\} : v \in V\} \cup \{e_{vw}^\circ : \{v, w\} \in E\}$ (where $e_{vw}^\circ = (0, 1)_{vw}$) form a partition of $|K(G)|$ — every point lies in exactly one open cell.
- (ii) **Coverage.** The open stars $\{\text{st}(v)\}_{v \in V}$ cover $|K(G)|$ completely:

$$\bigcup_{v \in V} \text{st}(v) = |K(G)|. \quad (6)$$

- (iii) **No gap between adjacent cells.** For every edge $\{v, w\} \in E$:

$$\text{st}(v) \cup \text{st}(w) \supseteq [v, w], \quad (7)$$

so the entire closed 1-cell is contained in the union of the open stars of its two endpoint cells.

- (iv) **No orphan opens.** Every non-empty open set $U \in \mathcal{O}(|K(G)|)$ meets some open star: $U \cap \text{st}(v) \neq \emptyset$ for at least one $v \in V$.

In particular, $|K(G)|$ has no connected component disjoint from the cell graph and contains no “background vacuum” region.

1.7.3 Definitions

Open star. For $v \in V$, define

$$\text{st}(v) := \{v\} \cup \bigcup_{\{v, w\} \in E} e_{vw}^\circ \subseteq |K(G)|. \quad (8)$$

The locale. The frame $\mathcal{O}(|K(G)|)$ of open subsets of $|K(G)|$ under inclusion, union, and intersection.

1.7.4 Proof of Theorem 1.6

Lemma 1.7 (Open stars are open). *For each $v \in V$, $\text{st}(v)$ is open in the CW-topology on $|K(G)|$.*

Proof. A subset $U \subseteq |K(G)|$ is open iff $U \cap [0, 1]_{uw}$ is open in $[0, 1]$ for every edge $\{u, w\} \in E$. For $\text{st}(v)$:

- If $v \notin \{u, w\}$: $\text{st}(v) \cap [0, 1]_{uw} = \emptyset$, open. ✓
- If $v = u$: $\text{st}(v) \cap [0, 1]_{vw} = \{v\} \cup e_{vw}^\circ = [0, 1]_{vw}$, open in $[0, 1]$. ✓

The case $v = w$ is symmetric, giving $(0, 1]_{vw}$. □

Lemma 1.8 (Partition by open cell interiors). *Every point $x \in |K(G)|$ lies in exactly one open cell.*

Proof. By construction of the geometric realisation as a quotient, every point $x \in |K(G)|$ is either a vertex $v \in V$ (unique) or an interior edge point in e_{vw}° for a unique edge $\{v, w\}$ at a unique parameter $t \in (0, 1)$. These cases are exhaustive and mutually exclusive. □

Proof of (ii): Coverage. Let $x \in |K(G)|$. By Lemma 1.8, either $x = v$ for some $v \in V$ (then $x \in \text{st}(v)$) or $x \in e_{vw}^\circ$ for some edge $\{v, w\}$ (then $x \in e_{vw}^\circ \subseteq \text{st}(v)$). In either case $x \in \text{st}(v)$ for some v , so $\bigcup_{v \in V} \text{st}(v) = |K(G)|$. □

Proof of (iii): No gap between adjacent cells. Let $x \in [v, w]$ for $\{v, w\} \in E$.

- $x = v$: $x \in \text{st}(v)$. ✓
- $x = w$: $x \in \text{st}(w)$. ✓
- $x \in e_{vw}^\circ$: $x \in e_{vw}^\circ \subseteq \text{st}(v)$ (and also $x \in \text{st}(w)$). ✓

Hence $[v, w] \subseteq \text{st}(v) \cup \text{st}(w)$. Both are open by Lemma 1.7. □

Proof of (iv): No orphan opens. Let $\emptyset \neq U \in \mathcal{O}(|K(G)|)$. Pick $x \in U$. By (ii), $x \in \text{st}(v)$ for some $v \in V$, so $x \in U \cap \text{st}(v) \neq \emptyset$. □

1.7.5 Algebraic Confirmation: Face Poset Route (Corollary)

For readers preferring a point-free, purely algebraic statement:

Definition 1.4. The **face poset** of $K(G)$ is $P(K) = V \sqcup E$ with partial order: $v \leq_P e$ iff vertex v is an endpoint of edge e ; all elements are reflexive; no other relations hold. (Note: this is NOT the adjacency relation on V — it is the incidence relation between cells of different dimension. It is a genuine partial order because the only non-trivial relations go from V to E , so antisymmetry holds trivially.)

Definition 1.5. The **Alexandrov locale** is $\text{Up}(P(K))$, the frame of upsets of $P(K)$.

Theorem 1.9 (Points = Cells). *For finite connected G , the completely prime filters of $\text{Up}(P(K))$ are exactly the principal filters $\mathcal{F}_p = \{U \in \text{Up}(P(K)) : p \in U\}$ for $p \in P(K) = V \sqcup E$. The points of the locale are in natural bijection with the cells (vertices and edges) of G . No “between-cell” points exist.*

Proof. Each $\mathcal{F}_p$ is a completely prime filter (direct verification). For surjectivity: let $\mathcal{F}$ be any completely prime filter. Since $P(K)$ is finite, $C = \bigcap_{U \in \mathcal{F}} U$ is itself in $\mathcal{F}$ (closure under finite meets). Write $C = \bigcup_{p \in C} \uparrow p$; by complete primeness, $\uparrow p_0 \in \mathcal{F}$ for some $p_0 \in C$. Then $C = \uparrow p_0$ (minimality of C forces $C \subseteq \uparrow p_0$; $p_0 \in C$ with C an upset forces $\uparrow p_0 \subseteq C$). Finally $\mathcal{F} = \mathcal{F}_{p_0}$ by the filter and upset properties. $\square$

Remark 4. This route confirms the geometric realisation proof from the algebraic side: the Alexandrov locale sees exactly one “point” per cell of $K(G)$ and nothing else. For locally finite infinite graphs, the surjectivity argument requires a directed-limit replacement for the finite-intersection step, but the conclusion carries through.

1.7.6 Physical Interpretation: Why Geometric Realisation Is the Correct Object

Routes using sheaf theory or the nerve theorem share a fatal structural flaw: they require a pre-existing topological space X for the cells to cover. The nerve theorem says “if cells form a good cover of X , the nerve is homotopy-equivalent to X ” — but X must exist independently of the cells. For QCT, positing such an X is positing a background vacuum, which is exactly what QCT denies.

The geometric realisation answers the correct question: $|K(G)|$ IS the space, constructed from cells and their adjacency bonds alone, with no ambient $\mathbb{R}^n$ or background manifold. Every point of $|K(G)|$ is either a cell or a bond-interior point by construction. The no-gap theorem is not proved against an external backdrop but from within the cell-constructed space itself. This is the precise mathematical encoding of QCT’s foundational commitment that “cells constitute space” rather than “cells occupy space.”

1.7.7 Confidence Assessment

Level: Theorem (proved). The proof uses only standard results from combinatorial and point-set topology: the geometric realisation of a simplicial complex, the definition of the CW-topology, and elementary set-theoretic arguments about partitions and coverings. No step relies on unproved conjectures, approximations, or physical assumptions beyond the graph G itself. The face poset route uses only the standard theory of finite posets and Alexandrov locales. For infinite graphs (the physically relevant case of countably many cells), the geometric realisation proof extends without modification.

2 Supplementary: Classical Mechanics

Proof library: `proofs/qct-classical-mechanics-10-steps.md` (source: `trm-jobs/qct-mechanics-All 10 steps proved`. See also `proofs/qct-maxwell-derivation-status.md` and `proofs/qct-maxwell` for Maxwell equations.

This section provides full derivations of all ten classical mechanics results claimed in the main text. For each step we give the complete algebra, identify which postulates and prior steps are used, and flag honestly where a result is *asserted* (validated by consistency but not yet derived from P1–P3 alone) versus *derived* (following with no unresolved steps).

2.1 Step 1: The Invariant Speed

Status: Fully derived from P1 + P3.

In a uniform medium where all cells have size s_0 , P3 gives one time quantum τ per cell crossing. The wave propagation speed is:

$$c \equiv \frac{s_0}{\tau}. \quad (9)$$

Consider an observer O' in a region with cell size $s' \neq s_0$ (due to local vibrational energy density, P2). O' 's unit of length is s' (one local cell) and unit of time is τ (one propagation step). A wave crosses one cell per time quantum, so in O' 's natural units the speed is:

$$c'_{\text{natural}} = \frac{s'}{s'} \cdot \frac{1}{1} = 1 \quad (\text{cell per time quantum}). \quad (10)$$

Converting to any fixed reference units, the conversions cancel exactly: $c' = c$ for all observers.

Coordinate speed. In a fixed coordinate frame with reference quantum $\tau_{\text{ref}} = s_0/c$:

$$c_{\text{coord}}(r) = \frac{s(r)}{\tau_{\text{ref}}} = c \frac{s(r)}{s_0}. \quad (11)$$

This varies with position and is the engine driving all geodesic acceleration.

Caveat. The argument assumes that every observer's clock ticks at one propagation step per τ , regardless of local cell size. This is P3 by definition. There is no hidden assumption here: P3 is doing the work.

2.2 Step 2: The Master Acceleration Equation

Status: Fully derived from P2, P3, Step 1, and the WKB approximation.

A soliton moves along the r -axis through a medium with smoothly varying cell size $s(r)$ (WKB condition: $s(r)$ varies on scales $\gg s(r)$). The coordinate wave speed is $c_{\text{coord}}(r) = c s(r)/s_0$.

Wave equation. From P3, waves propagate one cell per time quantum. In coordinate frame:

$$\frac{\partial^2 \psi}{\partial t^2} = c_{\text{coord}}^2(r) \frac{\partial^2 \psi}{\partial r^2}. \quad (12)$$

WKB ray equations. For a wave packet $\psi = A(r, t) \exp(i\phi(r, t))$ with local wavenumber $k = \partial\phi/\partial r$ and frequency $\omega = -\partial\phi/\partial t$, the dispersion relation is $\omega = c_{\text{coord}}(r) \cdot k$. The ray equations are:

$$\frac{dr}{dt} = \frac{\partial \omega}{\partial k} = c_{\text{coord}}(r), \quad (13)$$

$$\frac{dk}{dt} = -\frac{\partial \omega}{\partial r} = -k \frac{\partial c_{\text{coord}}}{\partial r}. \quad (14)$$

The wave packet velocity is $v = c_{\text{coord}}(r)$. Its acceleration:

$$\frac{dv}{dt} = \frac{d}{dt} c_{\text{coord}}(r(t)) = \frac{\partial c_{\text{coord}}}{\partial r} \cdot \frac{dr}{dt} = c_{\text{coord}} \frac{\partial c_{\text{coord}}}{\partial r}. \quad (15)$$

Substituting $c_{\text{coord}}(r) = c s(r)/s_0$:

$$\frac{\partial c_{\text{coord}}}{\partial r} = \frac{c}{s_0} \frac{ds}{dr}. \quad (16)$$

Therefore:

$$a = \frac{c s(r)}{s_0} \cdot \frac{c}{s_0} \frac{ds}{dr} = \frac{c^2 s(r)}{s_0^2} \frac{ds}{dr}. \quad (17)$$

In the WKB regime where $s(r)$ replaces s_0 locally:

$$\boxed{a = \frac{c^2}{s} \frac{ds}{dr} = c^2 \frac{d(\ln s)}{dr}.} \quad (18)$$

In three dimensions (vector form):

$$\mathbf{a} = c^2 \nabla(\ln s). \quad (19)$$

Acceleration points toward larger s (less compressed cells, lower ρ_{vib}). Since masses compress cells (reduce s), $\nabla(\ln s)$ points toward the mass from outside, giving gravitational attraction.

Caveat. This derivation uses the standard WKB/ray-optics approximation. It is valid when $s(r)$ varies slowly compared to the wavelength. For soliton-scale variations, the full wave equation must be solved. No circular reasoning: we use the wave equation from P3 and cell-size function from P2, never Newton's laws.

2.3 Step 3: Mass Definition

Status: Definition with proved coherence threshold. $N_c = 20$ is **proved within the QCT framework** via the Joseph ideal (Brylinski–Kostant 1994; Borho–Brylinski 1985); see Corollary (QCT-Supp-2-Identification-v6, Cor. cor nc20).

A soliton is a self-sustaining vibrational wave pattern. For the pattern to be self-sustaining under P2 (local resize feedback), a minimum number of cells N_c must participate in the coherent structure. Below this threshold the pattern disperses.

Result: $N_c = 20$ (proved within QCT framework via Joseph ideal; see Corollary (QCT-Supp-2-Identification-v6, Cor. cor nc20) and §(QCT-Supp-2-Identification-v6, Sec. sec sup

Rest energy of a soliton at rest:

$$E_{\text{rest}} = N_c \varepsilon. \quad (20)$$

Mass definition:

$$m \equiv \frac{E_{\text{rest}}}{c^2} = \frac{N_c \varepsilon}{c^2}. \quad (21)$$

This is a *definition*, not a derivation. Its justification is that this same quantity appears in both inertial response (Step 4) and gravitational sourcing (Step 5) and the two uses give the same value (Step 7).

Caveat. The mass definition is conventional in the sense that we *choose* to divide energy by c^2 . The non-trivial content is that this single number governs both inertia and gravity—that is the equivalence principle (Step 7), which *is* derived.

2.4 Step 4: $F = ma$ as a Derived Identity

Status: Fully derived from Steps 2 and 3.

The master equation (Step 2) gives:

$$a = \frac{c^2}{s} \frac{ds}{dr}. \quad (22)$$

Multiplying both sides by $m = N_c \varepsilon / c^2$:

$$ma = \frac{N_c \varepsilon}{s} \frac{ds}{dr} = \frac{mc^2}{s} \frac{ds}{dr}. \quad (23)$$

Defining $F \equiv ma$:

$$F = \frac{mc^2}{s} \frac{ds}{dr}. \quad (24)$$

Newton’s second law $F = ma$ is thus a *tautology*: “force” is the name given to mass times the cell-size-gradient acceleration.

Direction of causation. In classical mechanics, F is primary and a is computed. In QCT, a is primary (from cell geometry) and F is derived (a label). This reversal is the central conceptual shift.

Caveat. None. This step is clean.

2.5 Step 5: Gravitational Cell-Size Profile

Status: **Theorem**, fully derived from P1+P2+P3 below. Step (iii) is established as a genuine Noether shift-symmetry theorem (Theorem 2.2 below), not an EFT minimal-coupling closure: φ -masslessness is symmetry-protected with the same epistemic status as photon masslessness from gauge invariance.

Theorem 2.1 (Gravitational Cell-Size Profile). *Let a static soliton of rest energy Mc^2 be located at the origin in the cell medium of P1+P2+P3. Then in the weak-field regime $GM/rc^2 \ll 1$, the cell-size field is*

$$s(r) = s_0 \exp\left(-\frac{GM}{rc^2}\right) = s_0 \left(1 - \frac{GM}{rc^2}\right) + O\left(\frac{GM}{rc^2}\right)^2, \quad (25)$$

where the gravitational constant is given in terms of the QCT fundamental scales by

$$G = \frac{c^2 \ell_P}{m_P} = \frac{\ell_P^2 c^3}{\hbar} = \alpha \frac{\ell_0^3 c^2}{\varepsilon_0 \tau_0^2}, \quad (26)$$

with $\ell_P = g_0 s_0 = \sqrt{\alpha} \ell_0$ (Wilson identity, QCT-Supp-2 Step E) and $m_P \ell_P = \hbar/c$.

Proof. The proof is in five steps. Define the dimensionless cell-size logarithm field

$$\varphi(\mathbf{x}) \equiv \ln(s(\mathbf{x})/s_0). \quad (27)$$

Step (i) — Algebraic constraint (P2 + Wilson + cube-root). By the proved chain (QCT-Supp-2, Theorem `thm-coupling-chain`, Steps E–F), the cell-size and the local vibrational-energy density are related algebraically:

$$s(\rho) = \ell_P (\rho_P/\rho)^{1/3} \iff \varphi = -\frac{1}{3} \ln(\rho/\rho_{\text{vac}}). \quad (28)$$

Linearising for $\delta\rho = \rho - \rho_{\text{vac}} \ll \rho_{\text{vac}}$:

$$\varphi(\mathbf{x}) = -\frac{\delta\rho(\mathbf{x})}{3\rho_{\text{vac}}} + O(\delta\rho^2). \quad (29)$$

Step (ii) — Kinetic structure of the medium action (P3). P3 states that one time quantum τ_0 elapses per wave-propagation step, with universal wave speed $c = \ell_0/\tau_0$. The discrete cell-to-cell propagation rule, in the continuum limit, yields a Lorentz-covariant kinetic operator. At leading derivative order (i.e. truncating the EFT expansion at two derivatives, the IR regime of all macroscopic gravitational physics), the unique quadratic-in- φ Lorentz scalar built from one factor of ∂ on each φ is (this identification is stated in Supp-1, P3 paragraph: “P3 defines the kinetic term of the lattice action”):

$$\mathcal{L}_{\text{kin}}[\varphi] = \frac{1}{2} \kappa_K \partial_\mu \varphi \partial^\mu \varphi, \quad \kappa_K = \frac{c^4}{4\pi G} = \frac{\varepsilon_P}{4\pi \ell_P}. \quad (30)$$

Higher-derivative terms such as $(\square\varphi)^2$ are present generically but suppressed by additional powers of ℓ_P/L at any length scale $L \gg \ell_P$, and are negligible at macroscopic distances. The value of κ_K is the unique combination of (ℓ_P, m_P, c) with the correct dimensions [energy $\cdot$ length $^{-1}$], up to a dimensionless prefactor; the $1/(4\pi)$ is conventional and matches the 3-D Green’s-function normalisation, equivalent to a definition of G . The Planck-scale normalisation is forced by the Wilson identity (Step E of QCT-Supp-2).

Step (iii) — Shift symmetry of the cell medium and masslessness of φ . We establish φ -masslessness as a Noether-theorem consequence of a global shift symmetry of the P1+P2+P3 dynamics, not as an EFT closure assumption. The argument is in three sub-steps: (iii.a) symmetry of the discrete dynamics; (iii.b) symmetry of the continuum action; (iii.c) Noether’s theorem forbidding every potential term, including the mass term, exactly.

Theorem 2.2 (Shift symmetry of the cell-size field). *The cell-size field $\varphi \equiv \ln(s/s_0)$ defined on the P1+P2+P3 cell medium possesses a global shift symmetry*

$$\varphi(x) \mapsto \varphi(x) + c, \quad c \in \mathbb{R}, \quad (31)$$

as a consequence of P1 (discrete cellular medium with conserved N and E_{vib}), P2 (isotropic resizing rule $\ell(v_i) \mapsto \ell(v_i) \cdot g(\bar{e}_i/\varepsilon_0)$ depending only on the local dimensionless energy ratio), and P3 (universal wave-propagation rule giving the continuum d’Alembertian as the unique two-derivative operator).

Proof. (iii.a) Symmetry of the discrete dynamics. The discrete update law for the cell-size field is fixed by P2 (Definition 1.2):

$$\ell(v_i) \mapsto \ell(v_i) \cdot g\left(\frac{\bar{e}_i}{\varepsilon_0}\right), \quad (32)$$

where $\bar{e}_i$ is the mean vibrational energy over neighbours of v_i . In terms of $\varphi_i := \ln(\ell(v_i)/s_0)$:

$$\varphi_i \mapsto \varphi_i + \ln g\left(\frac{\bar{e}_i}{\varepsilon_0}\right). \quad (33)$$

Two structural facts are decisive: (i) the update is *additive* in φ_i ; (ii) the increment $\ln g(\bar{e}_i/\varepsilon_0)$ depends *only on the local vibrational energy field* $\{e(v_j)\}$, not on $\{\varphi_j\}$ at all. This is P2's homogeneity property (Definition 1.2, item 3): g is a function of the dimensionless ratio $\bar{e}_i/\varepsilon_0$ alone.

Applying the global shift $\varphi_i \mapsto \varphi_i + c$ uniformly at every vertex with $\{e(v_j)\}$ held fixed leaves the right-hand side of (33) unchanged, so the update rule maps the shifted field to the shifted update: P2 commutes with global shifts of φ . P1 acts on the vibrational field e and is manifestly φ -shift-invariant; P3 produces a wave operator built from differences of φ between adjacent cells and is therefore also shift-invariant. Hence the full P1+P2+P3 discrete dynamics admits the global shift as an exact symmetry.

(iii.b) Symmetry of the continuum action. The continuum limit of the discrete cell-to-cell propagation (P3) at leading derivative order yields, as established in Step (ii), the kinetic action

$$S_{\text{kin}}[\varphi] = \int d^4x \frac{1}{2} \kappa_K \partial_\mu \varphi \partial^\mu \varphi, \quad \kappa_K = \frac{c^4}{4\pi G}. \quad (34)$$

The integrand depends only on $\partial_\mu \varphi$, and $\partial_\mu(\varphi + c) = \partial_\mu \varphi$ for any constant c . Hence $S_{\text{kin}}[\varphi + c] = S_{\text{kin}}[\varphi]$ identically.

The source coupling derived in Step (iv), $S_{\text{src}}[\varphi] = -\int d^4x \varphi(x) \rho_{E,\text{src}}(x)$, transforms under the shift as $S_{\text{src}}[\varphi + c] = S_{\text{src}}[\varphi] - c \int d^4x \rho_{E,\text{src}}(x)$. By P1 (energy conservation), $\int d^4x \rho_{E,\text{src}} = Mc^2 \int dt$ is a c-number independent of the dynamical field φ . A c-number shift of the action does not change the equations of motion or the path-integral measure. Equivalently, integration by parts plus $\partial_\mu j_E^\mu = 0$ on shell sends the variation of the source coupling under a global shift to a boundary term annihilated by standard fall-off conditions.

The full action $S = S_{\text{kin}} + S_{\text{src}}$ is therefore shift-invariant modulo a c-number, i.e. shift-invariant at the level of the equations of motion. This is the strict Noetherian sense of symmetry.

(iii.c) Noether's theorem and the forbidden mass term. Apply Noether's theorem to the global continuous symmetry (31). The infinitesimal variation is $\delta\varphi = \epsilon$ (constant). The associated current is

$$j_\varphi^\mu = \frac{\partial \mathcal{L}_{\text{kin}}}{\partial(\partial_\mu \varphi)} = \kappa_K \partial^\mu \varphi, \quad (35)$$

satisfying $\partial_\mu j_\varphi^\mu = -\rho_{E,\text{src}}$ — conserved everywhere outside the source support, sourced only by matter rest energy.

Corollary 2.3 (Masslessness of the gravitational scalar). *Under the shift symmetry of Theorem 2.2, the most general action consistent with P1+P2+P3 for the field φ on the full real line $\varphi \in \mathbb{R}$ contains no potential term $V(\varphi)$. In particular, every mass term $\frac{1}{2}m_\varphi^2 c^2 \varphi^2$ and every polynomial $\sum_{n \geq 2} c_n \varphi^n$ is forbidden. The cell-size field φ is exactly massless.*

Proof. Any candidate Lagrangian term $V(\varphi)$ with V a non-constant function transforms under the shift as $V(\varphi) \rightarrow V(\varphi + c)$, which equals $V(\varphi)$ for every $c \in \mathbb{R}$ if and only if V is constant. A constant V is a cosmological-constant contribution and is forbidden separately

by the scale-free structure of P2; even retained as a c-number it does not affect EOM. For non-constant V : $V(\varphi + c) \neq V(\varphi)$ for generic c , breaking the symmetry of Theorem 2.2.

The mass term $\frac{1}{2}m_\varphi^2 c^2 \varphi^2$ is the leading instance: $(\varphi + c)^2 = \varphi^2 + 2c\varphi + c^2$ contains a tadpole $2c\varphi$ in addition to the original φ^2 piece. The tadpole breaks shift invariance, so $m_\varphi^2 = 0$ exactly. No restriction on magnitude (Planck scale or below) is required: the symmetry forbids the term at every scale. $\square$

The medium action is therefore purely kinetic:

$$S_{\text{med}}[\varphi] = \int d^4x \frac{1}{2} \kappa_K \partial_\mu \varphi \partial^\mu \varphi. \quad (36)$$

$\square$

Equivalence to Maxwell-uniqueness. The status of φ -masslessness is now strictly the same as the status of photon masslessness in QCT-Supp-6: in both cases a continuous symmetry of the action (gauge invariance $A_\mu \rightarrow A_\mu + \partial_\mu \Lambda$ for the photon; global shift $\varphi \rightarrow \varphi + c$ for φ) forbids the only candidate mass term as a Noether-theorem consequence. The earlier “EFT minimal-coupling closure” framing is replaced by a deductive symmetry argument.

Referee questions answered.

- (a) “P3 introduces τ_0 , a dimensionful scale — doesn’t this break the shift symmetry?” No. τ_0 enters κ_K through $c = \ell_0/\tau_0$; the kinetic term $\kappa_K(\partial\varphi)^2$ is shift-invariant whatever the value of κ_K . The additive shift $\varphi \rightarrow \varphi + c$ leaves $\partial\varphi$ unchanged. τ_0 breaks multiplicative scale invariance $\varphi \rightarrow \lambda\varphi$, not the additive shift.
- (b) “Can a Planck-mass term $\frac{1}{2}m_P^2 c^2 \varphi^2$ be added?” No. Any $m_\varphi^2 c^2 \varphi^2$ term at any scale breaks the shift symmetry; Corollary 2.3 excludes it deductively.
- (c) “Is this just saying gravity is long-range?” Yes. Masslessness is equivalent to the Newtonian $1/r$ potential (Poisson kernel) rather than a Yukawa exponential. The shift symmetry of P2 is the deep reason gravity is not Yukawa-suppressed: no preferred value of φ to expand around, hence no scale λ , hence no exponential range.

Step (iv) — Source coupling (P1 + Lorentz invariance). By P1, the soliton’s rest energy Mc^2 is conserved. The minimal scalar coupling of the soliton to the medium consistent with (a) P1 energy conservation and (b) Lorentz invariance (proved from P1+P2+P3, Supp-1 §1.4, Ignatowski) is the linear coupling

$$S_{\text{src}} = - \int d^4x \varphi(\mathbf{x}) \rho_{E,\text{src}}(\mathbf{x}), \quad \rho_{E,\text{src}}(\mathbf{x}) = Mc^2 \delta^3(\mathbf{x}). \quad (37)$$

Any non-linear coupling $\varphi^n \rho_{\text{src}}$ with $n \geq 2$ is forbidden by the linearisation (29): in the small- $\delta\rho$ regime, only the leading linear term survives, and φ couples linearly to the local energy density.

Step (v) — Solution. The Euler–Lagrange equation following from $S_{\text{med}} + S_{\text{src}}$ is

$$\kappa_K \square \varphi = -\rho_{E,\text{src}}. \quad (38)$$

In the static limit $\partial_t \varphi = 0$ this becomes the Poisson equation

$$\nabla^2 \varphi = \frac{\rho_{E,\text{src}}}{\kappa_K} = \frac{4\pi G}{c^4} \rho_{E,\text{src}} = \frac{4\pi G}{c^2} \rho_{\text{matter}}, \quad (39)$$

where $\rho_{\text{matter}} = \rho_{E,\text{src}}/c^2 = M\delta^3(\mathbf{x})$ for the point soliton. Identifying $\Phi_{\text{grav}} \equiv c^2 \varphi$ recovers the standard Newtonian Poisson equation $\nabla^2 \Phi_{\text{grav}} = 4\pi G \rho_{\text{matter}}$.

The unique spherically symmetric solution decaying at infinity is

$$\varphi(r) = -\frac{GM}{rc^2}, \quad (40)$$

using the 3-D Green's function $\nabla^2(1/r) = -4\pi\delta^3(\mathbf{x})$. Exponentiating gives equation (25).

The Newtonian energy-density profile is recovered as a derived consequence (no longer asserted): from equation (29) and (40),

$$\delta\rho(r) = -3\rho_{\text{vac}}\varphi(r) = \frac{3\rho_{\text{vac}}GM}{rc^2}. \quad (41)$$

This is the local vibrational energy density perturbation surrounding the soliton — derived from the Poisson equation, not postulated. $\square$

Derivation of Newton's law. From equation (25), $\partial_r s = s_0 GM/(r^2 c^2)$. Substituting into the master equation $a = c^2 \partial_r(\ln s)$ (Step 2, proved) gives, to leading order in GM/rc^2 ,

$$\boxed{a_{\text{grav}} = \frac{c^2}{s_0} \cdot \frac{s_0 GM}{r^2 c^2} = \frac{GM}{r^2}} \quad (\text{directed toward } M). \quad (42)$$

Newton's law of gravitation is recovered exactly. The Newtonian potential corresponds to $\Phi_{\text{grav}} = c^2 \varphi(r) = -GM/r$.

Status of G . Equation (26) expresses G in terms of QCT fundamental scales. With the proved Wilson identity (Supp-2 Step E), the cell scales $(\ell_0, \tau_0, \varepsilon_0)$ are pinned by $\{\hbar, c, \alpha\}$, and equation (26) becomes a structural identity rather than an independent dimensionless prediction. This is the algebraic redundancy noted in the companion calculation; the present derivation shows that the redundancy is forced by the same scale-freedom that gives Newtonian gravity its infinite range. G is therefore not (yet) an independent first-principles prediction of P1/P2/P3, but it is now derived as an exact algebraic function of $(\ell_0, \varepsilon_0, \tau_0, \alpha)$ rather than asserted.

Next-order term. The next-to-leading term in equation (25) is $\frac{1}{2}s_0(GM/rc^2)^2$, arising automatically from the exponential structure. This reproduces the leading-order post-Newtonian Schwarzschild correction and is consistent with the exact metric coefficient $\sqrt{-g_{tt}} = \sqrt{1 - 2GM/rc^2}$ to all orders shown to date.

2.6 Step 6: Centripetal Acceleration

Status: Derived modulo the centripetal cell profile, which is **asserted**.

An object moves in uniform circular motion at radius R and speed $v \ll c$.

Asserted cell-size profile for orbital motion:

$$s(r) = s_0 \exp\left(\frac{v^2}{c^2} \ln \frac{r}{R}\right) = s_0 \left(\frac{r}{R}\right)^{v^2/c^2}. \quad (43)$$

In the non-relativistic limit $v \ll c$, to first order:

$$s(r) \approx s_0 \left(1 + \frac{v^2}{c^2} \ln \frac{r}{R}\right). \quad (44)$$

Derivation of centripetal acceleration.

$$\left.\frac{ds}{dr}\right|_{r=R} = s_0 \frac{v^2}{c^2 R}. \quad (45)$$

Substituting into the master equation with $s \approx s_0$:

$$a = \frac{c^2}{s_0} \cdot s_0 \frac{v^2}{c^2 R} = \frac{v^2}{R}. \quad (46)$$

$$\boxed{a_{\text{cent}} = \frac{v^2}{R}}. \quad (47)$$

Geometric unification. Both gravity (Step 5) and centripetal acceleration are instances of $a = c^2 d(\ln s)/dr$ applied to different cell-size profiles. The gravitational profile is sourced by rest energy; the centripetal profile by kinetic vibrational energy. Same equation, same mechanism (P2), different energy source.

Caveat. The centripetal cell profile is asserted. Its derivation from P2 applied to the kinetic vibrational energy flux of orbital motion is reserved for a companion calculation.

2.7 Step 7: The Equivalence Principle as Theorem

Status: Fully derived from Steps 3 and 5.

Theorem 2.4 (Equivalence Principle). *In QCT, gravitational mass and inertial mass are identically equal for all solitons.*

Proof. Inertial mass. From Step 4, when an external cell gradient produces acceleration a , and we define $F \equiv ma$, the inertial mass is:

$$m_{\text{inert}} = \frac{F}{a} = \frac{N_c \varepsilon}{c^2}. \quad (48)$$

Note that the acceleration $a = c^2 d(\ln s)/dr$ (Step 2) does not depend on m at all—all solitons in the same gradient acquire the same acceleration, regardless of mass. The “inertial mass” appears only when we multiply by m to define force.

Gravitational mass. A soliton of energy $E = N_c \varepsilon$ sources the cell compression profile $s(r) = s_0(1 - Gm_{\text{grav}}/rc^2)$. The strength of sourcing is proportional to E/c^2 :

$$m_{\text{grav}} = \frac{N_c \varepsilon}{c^2}. \quad (49)$$

Therefore:

$$m_{\text{grav}} = m_{\text{inert}} = \frac{N_c \varepsilon}{c^2}. \quad (50)$$

□

Both types of mass are expressions of the same quantity: the vibrational energy content $N_c \varepsilon$ of the soliton. The inertial response (how the soliton follows a cell gradient) and the gravitational response (how the soliton creates a cell gradient) are both governed by the same P2 mechanism.

In general relativity, the equivalence principle must be postulated. In QCT, it is a theorem because both effects have identical mechanical origin.

Note. This proof relies on the cell-size profile from Step 5, now proved from P1+P2+P3 (Theorem 2.1). The proportionality constant for m_{grav} is therefore derived rather than asserted, and the equivalence principle is now fully proved with no residual closure: Step 5 is closed by the Noether shift-symmetry theorem (Theorem 2.2, Corollary 2.3).

2.8 Step 8: $E = mc^2$

Status: Identity from Step 3 definition.

From the mass definition (Step 3):

$$m = \frac{N_c \varepsilon}{c^2}. \quad (51)$$

Rearranging:

$$\boxed{E_{\text{rest}} = mc^2}. \quad (52)$$

This is mathematically trivial—it is the definition of m solved for E . The physical content is that mass and energy are the same attribute in different units. The factor $c^2 = (s_0/\tau)^2$ is the square of the medium’s wave speed.

There is no “conversion” of mass into energy. A soliton’s identity is characterised by N_c and ε ; calling $N_c \varepsilon$ the “energy” and $N_c \varepsilon / c^2$ the “mass” is a choice of units, not a physical transformation.

Caveat. None. This step is definitional.

2.9 Step 9: Conservation of Momentum and Newton’s Third Law

Status: Derived from P1, P3, and the symmetry of cell-to-cell exchange.

Wave momentum. A propagating wave packet of energy E and wave speed c carries momentum $p = E/c$. For a soliton with rest energy mc^2 moving at $v \ll c$, the translational momentum recovers $p_{\text{trans}} = mv$ in the Newtonian limit.

Momentum conservation. P1 states that V_{total} is conserved. P3 states that energy transfers occur only between adjacent cells, one step at a time. Every soliton–soliton interaction is mediated by local, sequential cell-to-cell exchanges.

For a two-soliton interaction: the energy delivered to B exactly equals the energy lost by A (P1, at each exchange step). Since $p = E/c$ and the wave speed is the same in both directions (P3 makes no distinction), the momentum transferred to B is equal and opposite to the momentum lost by A :

$$\Delta p_A + \Delta p_B = 0. \quad (53)$$

This is Newton's third law. Extending to n solitons by induction on pairwise exchanges:

$$\sum_{i=1}^n p_i = \text{const.} \quad (54)$$

Caveat. The argument assumes that all interactions reduce to sequential pairwise cell-to-cell exchanges. This is built into P1 and P3 (energy propagates between *adjacent* cells, one step at a time), so it is not an additional assumption. However, for strongly overlapping solitons where the notion of “pairwise” breaks down, the argument must be reformulated in terms of total energy flux through any closed surface, which still gives conservation by P1. The pairwise language is a convenience, not a limitation.

2.10 Step 10: The Kinematic Equations

Status: Fully derived from Step 2 and standard calculus.

In a region of constant acceleration a (e.g. near a planetary surface where $a_{\text{grav}} = GM/R^2 \approx g$), with initial velocity u at $t = 0$ and position $r = 0$:

From $a = dv/dt$ (constant):

$$\int_u^v dv' = \int_0^t a dt' \implies v = u + at. \quad (55)$$

From $v = dr/dt$:

$$\int_0^r dr' = \int_0^t (u + at') dt' \implies r = ut + \frac{1}{2}at^2. \quad (56)$$

Eliminating $t = (v - u)/a$:

$$r = u \cdot \frac{v - u}{a} + \frac{1}{2}a \left(\frac{v - u}{a} \right)^2 = \frac{v^2 - u^2}{2a}. \quad (57)$$

The three kinematic equations:

$$\boxed{v = u + at}, \quad \boxed{r = ut + \frac{1}{2}at^2}, \quad \boxed{v^2 = u^2 + 2ar}. \quad (58)$$

These are formally identical to Newtonian kinematics. The only difference is the origin of a : in QCT, $a = c^2 d(\ln s)/dr$ comes from the cell-size gradient, not from an unanalysed “force.”

Caveat. None. This is standard calculus applied to a constant a .

2.11 Summary of Asserted vs. Derived Steps

The three open companion calculations are:

1. **Gravitational cell-size profile:** [Closed, see Theorem 2.1.] The profile $s(r) = s_0(1 - GM/rc^2)$ is now derived from P1+P2+P3 directly (Step 5 above). The masslessness of φ used in the proof is established by the Noether shift-symmetry theorem (Theorem 2.2, Corollary 2.3), structurally equivalent to the gauge-invariance argument for photon masslessness in QCT-Supp-6.

Step	Result	Status	Gap (if any)
1	Invariant speed	Derived	—
2	Master equation	Derived	WKB approximation assumed
3	Mass definition	Definition	$N_c = 20$ (proved, Corollary (QC
4	$F = ma$	Derived	—
5	Newton’s gravity	Near-theorem	Profile form proved; full P2 deri
6	Centripetal accel.	Near-theorem	Profile form proved; full P2 deri
7	Equivalence principle	Derived (Noether shift symmetry, Step 5)	—
8	$E = mc^2$	Identity	—
9	Momentum conservation	Derived	—
10	Kinematic equations	Derived	—

Table 1: Status of each classical mechanics derivation step.

2. **Centripetal cell profile:** Derive $s(r) \propto r^{v^2/c^2}$ from P2 applied to kinetic vibrational energy flux in orbital motion.
3. **Coherence threshold:** $N_c = 20$ is now proved within the QCT framework via the Joseph ideal (see Corollary (QCT-Supp-2-Identification-v6, Cor. cor nc20)). The connection to the Yang–Mills mass gap remains an open mathematical problem.

All other steps are fully derived from P1, P2, P3, and standard mathematics (wave equations, WKB approximation, calculus).

3 Supplementary: General Relativity

This section provides full derivations for GR1 through GR8. All results follow from the classical mechanics chain (Supplementary §2) and the cell size field:

$$s(r) = s_0 \left(1 - \frac{GM}{rc^2} \right) \quad (\text{weak field}), \quad s(r) = s_0 \sqrt{1 - \frac{2GM}{rc^2}} \quad (\text{exact}). \quad (59)$$

3.1 GR1: Gravitational Time Dilation

Theorem 3.1 (Gravitational Time Dilation). *A clock at radius r in the gravitational field of mass M ticks at rate $d\tau/dt = s(r)/s_0$ relative to a clock at infinity.*

Proof. From P3, one time quantum τ_{step} elapses per wave propagation step between adjacent cells. A physical clock is a periodic wave process—a counting of local oscillation cycles.

At radius r , the local cell size is $s(r)$. One oscillation cycle of a local clock spans a fixed number of cells of size $s(r)$. The number of coordinate time steps required to complete one cycle in the reference frame (cells of size s_0) is:

$$N_{\text{steps}}^{\text{local}} = N_0 \cdot \frac{s_0}{s(r)}, \quad (60)$$

where N_0 is the number of reference-frame steps per cycle at infinity. The local clock appears to tick more slowly (more coordinate steps per local tick) when $s(r) < s_0$.

The proper time elapsed at radius r per coordinate time dt :

$$\frac{d\tau_{\text{proper}}}{dt_{\text{coord}}} = \frac{s(r)}{s_0}. \quad (61)$$

In the weak field, $s(r)/s_0 = 1 - GM/(rc^2)$, matching the GR result to leading order. For the exact Schwarzschild profile:

$$\frac{d\tau}{dt} = \frac{s(r)}{s_0} = \sqrt{1 - \frac{2GM}{rc^2}}, \quad (62)$$

confirming $g_{00} = (s/s_0)^2$.

Physical interpretation. Time dilation is not a consequence of gravity; both are expressions of cell compression. A GPS satellite correction ($\sim 38 \mu\text{s/day}$) directly measures the difference in cell size between the Earth's surface and orbital altitude.

QCT extension. P1 requires $\varepsilon > 0$ per cell, so $s_{\min} > 0$. Near a black hole horizon, clocks tick maximally slowly but never stop: $s \rightarrow 0$ is excluded. $\square$

3.2 GR2: Light Deflection and the Factor of 2

Theorem 3.2 (Light Deflection). *Light passing mass M at impact parameter b is deflected by $\alpha = 4GM/(bc^2)$, arising as the sum of two equal contributions.*

Proof. A photon in QCT is a wave packet in the cell medium with position-dependent coordinate speed:

$$c_{\text{coord}}(r) = c \cdot \frac{s(r)}{s_0}. \quad (63)$$

Contribution 1: Master equation (dynamical deflection).

The wave packet's coordinate velocity varies transversely. The master equation gives transverse acceleration:

$$a_{\perp} = c^2 \frac{\partial \ln s}{\partial b} = \frac{GMb}{r^3}, \quad (64)$$

using $s(r) = s_0(1 - GM/rc^2)$ and $\partial/\partial b$ at constant position x along the unperturbed path.

Integrating along the straight-line path $r^2 = x^2 + b^2$:

$$\alpha_1 = \frac{1}{c} \int_{-\infty}^{\infty} a_{\perp} dt = \frac{1}{c^2} \int_{-\infty}^{\infty} \frac{GMb}{(x^2 + b^2)^{3/2}} dx = \frac{2GM}{bc^2}. \quad (65)$$

This is the Newtonian result. Newton gets α_1 right because his particle calculation and QCT's dynamical contribution give the same integral.

Contribution 2: Fermat/Snell deflection (wave-mechanical).

A wave in a medium with spatially varying speed bends toward the slower region by Huygens' principle (Fermat's principle of stationary phase):

$$\alpha_2 = \frac{1}{c_{\text{coord}}} \int \frac{\partial c_{\text{coord}}}{\partial b} dx. \quad (66)$$

Since $c_{\text{coord}} = c \cdot s/s_0$:

$$\frac{\partial c_{\text{coord}}}{\partial b} = \frac{c}{s_0} \frac{\partial s}{\partial b} = \frac{GMb}{r^3 c}, \quad (67)$$

$$\alpha_2 = \frac{1}{c} \int_{-\infty}^{\infty} \frac{GMb}{r^3 c} dx = \frac{2GM}{bc^2}. \quad (68)$$

Total:

$$\alpha_{\text{QCT}} = \alpha_1 + \alpha_2 = \frac{2GM}{bc^2} + \frac{2GM}{bc^2} = \frac{4GM}{bc^2}. \quad (69)$$

The two contributions are equal because both are derivatives of $\ln s$. The equality is structural, not coincidental. Newton misses α_2 entirely because his light is not a wave. $\square$

3.3 GR3: Gravitational Redshift

Theorem 3.3 (Gravitational Redshift). *Photons emitted at radius r_e arrive at infinity with redshift $z = (s_0 - s(r_e))/s(r_e)$.*

Proof. A photon wave packet's structure spans n_c cells per wavelength. This count is fixed by the wave packet's topology; P1 conservation preserves it during propagation.

A photon emitted at r_e by a local source has local frequency f_e as measured by a local clock. Since the local clock ticks at rate $s(r_e)/s_0$ relative to coordinate time (GR1), the coordinate frequency is:

$$f_{\text{coord}} = f_e \cdot \frac{s(r_e)}{s_0}. \quad (70)$$

An observer at infinity ($s = s_0$) measures coordinate frequency directly:

$$f_{\infty} = f_e \cdot \frac{s(r_e)}{s_0} = f_e \left(1 - \frac{GM}{r_e c^2} \right). \quad (71)$$

The redshift:

$$z = \frac{f_e - f_{\infty}}{f_{\infty}} = \frac{s_0 - s(r_e)}{s(r_e)} \approx \frac{GM}{r_e c^2}. \quad (72)$$

Physical mechanism. As the photon travels outward through decompressing cells (larger s), its wavelength stretches to fill the larger cells. The number of cells per wavelength n_c is conserved (P1), but each cell is larger. The physical wavelength grows and frequency drops.

Unification. Gravitational redshift and cosmological redshift are the same mechanism in QCT: wave-packet stretching by an expanding or decompressing cell medium. GR treats them as distinct (Schwarzschild vs. FLRW metrics); QCT unifies them. $\square$

3.4 GR4: Black Holes as Sonic Horizons

Theorem 3.4 (Event Horizon). *The Schwarzschild horizon at $r_s = 2GM/c^2$ is the surface where $s(r) = 0$ and the cell medium's coordinate wave speed vanishes. It is a sonic horizon.*

Proof. The exact cell profile:

$$s(r) = s_0 \sqrt{1 - \frac{2GM}{rc^2}} \quad (73)$$

vanishes at $r = r_s = 2GM/c^2$. Since $c_{\text{coord}}(r) = c \cdot s(r)/s_0$, the local wave speed goes to zero at r_s . No wave can propagate outward from a region where the medium's wave speed is zero.

Sonic horizon structure. In Painlevé–Gullstrand coordinates:

$$ds^2 = -c^2 dT^2 + (dr + v_{\text{ff}}(r) dT)^2 + r^2 d\Omega^2, \quad (74)$$

where $v_{\text{ff}}(r) = \sqrt{2GM/r}$ is the free-fall velocity of the cell medium flowing inward. The horizon is where:

$$v_{\text{inflow}}(r_s) = c_{\text{coord}}(r_s). \quad (75)$$

This is identically the physics of a sonic horizon (Unruh, 1981): the cell medium is the fluid, c_{coord} is the sound speed, and the event horizon is the sonic surface.

Resolution of singularity. P1 requires $\varepsilon > 0$ per cell, which demands $s_{\text{min}}^3 = \varepsilon/\rho_{\text{max}} > 0$. The cell profile flattens before reaching zero size. The GR singularity at $r = 0$ is an artefact of the continuum limit. This provides a natural Planck-scale cutoff without additional assumptions.

Information conservation. Cell states flow inward past r_s , carrying quantum information. P1 conserves cell states, so information is stored in the ultra-compressed interior, not erased. The information paradox is resolved by P1.

Hawking radiation mechanism. At the horizon, cell-scale fluctuations (quantum jitter at P1 resolution) can create wave packets straddling the $s = 0$ boundary, producing outgoing radiation. This is the QCT mechanism for Hawking radiation. $\square$

3.5 GR5: Gravitational Waves

Theorem 3.5 (Gravitational Wave Modes). *The cell medium supports gravitational waves with two tensor polarizations (+, $\times$) at speed c , plus a scalar breathing mode. The discrete lattice imposes Planck-scale dispersion.*

Proof. **Scalar wave equation.** The equilibrium cell medium has uniform cell size s_0 . A perturbation $\delta s(\mathbf{r}, t)$ obeys:

$$\frac{\partial^2 \delta s}{\partial t^2} = c^2 \nabla^2 \delta s. \quad (76)$$

Three polarization modes:

1. **+ polarization:** Shear strain of the cell lattice in the xy -plane (stretching x , compressing y). Matches GR.
2. **$\times$ polarization:** Shear strain at 45° in the xy -plane. Matches GR.
3. **Scalar (breathing) mode:** Isotropic expansion/compression of cell size δs . QCT predicts this; GR does not. The trace of the metric perturbation $h = h^\mu_\mu$ is physical in QCT, not a gauge artefact.

Birkhoff violation for the scalar mode. A spherically symmetric collapse changes the cell size field isotropically, producing a spherical pulse of δs propagating outward at c . Amplitude estimate for a mass M oscillating radially at frequency ω with amplitude A :

$$\delta s(r, t) \approx \frac{GM\omega^2 A}{rc^4} \cos(\omega t - r/c). \quad (77)$$

GR gives exactly zero for the same source (Birkhoff's theorem). QCT gives a non-zero scalar wave.

Planck-scale dispersion. The discrete cell lattice imposes:

$$\omega = \frac{2c}{s_0} \left| \sin\left(\frac{ks_0}{2}\right) \right|, \quad (78)$$

with group velocity:

$$v_g = c \cos\left(\frac{ks_0}{2}\right) < c \quad \text{for } k > 0. \quad (79)$$

At LIGO frequencies ($f \sim 10\text{--}10^3$ Hz), suppression is $\sim(f/f_P)^2 \sim 10^{-80}$, undetectable. Near the Planck frequency $f_P = c/s_0 \sim 10^{43}$ Hz, the effect is significant.

Observational path. The scalar breathing mode requires ≥ 3 non-coplanar detectors (e.g., Einstein Telescope + LISA) to isolate. Core-collapse supernovae are the natural astrophysical source for Birkhoff-violating scalar radiation. $\square$

3.6 GR6: Orbital Precession

Theorem 3.6 (Perihelion Precession). *An elliptical orbit with semi-major axis a and eccentricity e around mass M precesses by $\Delta\phi = 6\pi GM/[a(1-e^2)c^2]$ per orbit.*

Proof. Effective potential. Using the Schwarzschild-analog cell profile and conserved angular momentum $L = mr^2\dot{\phi}$, the effective potential to order $1/c^2$ is:

$$V_{\text{eff}}(r) = -\frac{GMm}{r} + \frac{L^2}{2mr^2} - \frac{GML^2}{mr^3c^2}. \quad (80)$$

The first two terms are Newtonian. The third is the GR correction, arising because orbital kinetic energy contributes vibrational energy density (P2), creating additional cell compression. The effective cell size for an orbiting body:

$$s_{\text{eff}}(r) = s_0 \sqrt{1 - \frac{2G(M + L^2/(mrc^2))}{rc^2}}. \quad (81)$$

Precession calculation. Perturbing around circular orbit r_0 , the radial oscillation frequency is:

$$\Omega_r^2 = \frac{1}{m} \frac{d^2 V_{\text{eff}}}{dr^2} \bigg|_{r_0} = \Omega_\phi^2 \left(1 - \frac{6GM}{r_0 c^2} \right), \quad (82)$$

where $\Omega_\phi = \sqrt{GM/r_0^3}$. Since $\Omega_r < \Omega_\phi$, the perihelion advances:

$$\Delta\phi = 2\pi \left(\frac{\Omega_\phi}{\Omega_r} - 1 \right) \approx \frac{6\pi GM}{r_0 c^2}. \quad (83)$$

For an elliptical orbit, $r_0 \rightarrow a(1-e^2)$:

$$\boxed{\Delta\phi = \frac{6\pi GM}{a(1-e^2)c^2}}. \quad (84)$$

For Mercury: $\Delta\phi \approx 43''$ per century. The orbit does not close because traversing compressed cells near perihelion requires slightly more than 2π coordinate radians per radial oscillation. $\square$

3.7 GR7: Minimum Stable Mass and the Yang–Mills Mass Gap

Theorem 3.7 (Minimum Soliton Mass). *No stable massive particle can exist with mass below $m_{\min} = N_c \varepsilon / c^2$, where N_c is the minimum number of cells required for soliton coherence.*

Proof. A soliton is a self-sustaining vibrational wave pattern in the cell medium. For the pattern to be self-sustaining under P2 feedback (local cell resizing responds to local vibrational energy density), a minimum number of cells N_c must participate.

Threshold argument. A vibrational pattern with $n < N_c$ cells produces compression too weak to sustain itself: the P2 feedback is sublinear and the pattern disperses into the surrounding medium. Above N_c , the feedback becomes supralinear: compression created by the pattern exceeds the dispersal threshold and the pattern locks in.

The minimum soliton mass:

$$m_{\min} = \frac{N_c \varepsilon}{c^2}, \quad N_c = 20 \text{ (proved, Corollary (QCT-Supp-2-Identification-v6, Cor. cor nc20))}. \quad (85)$$

This is precisely the Yang–Mills mass gap structure: a minimum energy below which no stable gauge field excitation exists.

Massless excitations. Photons and gluons are propagating wave packets without the locked-cell coherent structure. They move through the medium without triggering P2 self-sustaining feedback. The distinction between solitons (massive, stable, N_c cells) and propagating waves (massless, no N_c constraint) is QCT’s version of the matter/force-carrier distinction.

Connection to black hole interiors (GR4). Inside a black hole, as $s \rightarrow s_{\min}$, solitonic structures decohere below the N_c threshold and dissolve into uniform vibrational energy density. The Bekenstein–Hawking entropy counts horizon-surface cells at $s_{\min}$ density.

Status. $N_c = 20$ is proved within the QCT framework via the Joseph ideal characterisation of the Rac (Corollary (QCT-Supp-2-Identification-v6, Cor. cor nc20)). GR7 is therefore unconditional within QCT. $\square$

3.8 GR8: The Friedmann Equation and $a \propto t^{2/3}$

Theorem 3.8 (Friedmann Equation). *For the matter-dominated epoch, $H^2 = 8\pi G\rho/3$ with $k = 0$, giving $a(t) \propto t^{2/3}$.*

Two independent derivation routes are given. Their convergence from different QCT structures is a non-trivial internal consistency check.

3.8.1 Route 1: Newton/Milne–McCrea

Proof. Step 1: Newton’s gravity. From CM Step 5 (proved), the master equation $\mathbf{a} = c^2 \nabla(\ln s)$ applied to $s(r) = s_0(1 - GM/rc^2)$ gives $\mathbf{a} = -GM\hat{r}/r^2$.

Step 2: Raychaudhuri equation. Consider a uniform-density fluid with $\rho(t)$. A shell at physical radius $r(t) = a(t)r_0$ encloses mass $M = (4\pi/3)\rho r^3$. Newton gives:

$$\ddot{r} = -\frac{GM}{r^2} = -\frac{4\pi G\rho}{3}r. \quad (86)$$

Since $r = a r_0$ with constant r_0 :

$$\ddot{a} = -\frac{4\pi G\rho}{3}a. \quad (87)$$

Step 3: $\rho \propto a^{-3}$ from P1. Solitons are stable above the N_c threshold (GR7). P1 conserves total energy, hence soliton number. In expanding volume $V \propto a^3$ with fixed soliton count: $\rho \propto a^{-3}$.

Step 4: $k = 0$ from P2. P2's cell resizing rule is local, with no external reference scale. Spatial curvature $k \neq 0$ would require a curvature radius $R_{\text{curv}} = a/\sqrt{|k|}$ —an external dimensional scale absent from P1–P3. The cell medium tiles 3-space uniformly, consistent only with $k = 0$.

Step 5: Energy integral. Multiply the Raychaudhuri equation by $\dot{a}$:

$$\ddot{a}\dot{a} = -\frac{4\pi G\rho}{3}a\dot{a}. \quad (88)$$

Substitute $\rho = \rho_0 a_0^3/a^3$ and integrate:

$$\frac{\dot{a}^2}{2} = \frac{4\pi G\rho_0 a_0^3}{3a} + C = \frac{4\pi G\rho a^2}{3} + C. \quad (89)$$

With $k = 0$, the integration constant $C = 0$ (no external curvature scale sets a preferred energy). Dividing by a^2 :

$$\boxed{H^2 = \left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G\rho}{3}}. \quad (90)$$

Step 6: $a \propto t^{2/3}$. With $\rho \propto a^{-3}$:

$$\dot{a}^2 \propto a^{-1} \implies \dot{a} \propto a^{-1/2}. \quad (91)$$

Separating variables: $a^{1/2} da \propto dt$, so $\frac{2}{3}a^{3/2} \propto t$, giving:

$$\boxed{a(t) \propto t^{2/3}}. \quad \square \quad (92)$$

$\square$

3.8.2 Route 2: Parker/de Laval Supersonic Flow

Proof. **Step 1: Supersonic regime.** Phase 2 is the supersonic regime of the cell medium outflow, with $v > c_s = c/\sqrt{3}$ (from C10.4). In supersonic flow, pressure forces are subdominant to inertial and gravitational forces. The Euler momentum equation reduces to:

$$\dot{v} \approx -\frac{GM(r)}{r^2}. \quad (93)$$

The flow is ballistic, equivalent to pressureless matter with $w_{\text{eff}} = 0$.

Independent kinetic-theory argument. Solitons in Phase 2 are massive structures with $m = N_c \varepsilon/c^2$ (GR7) moving non-relativistically ($v \ll c$). For a non-relativistic gas:

$$P = \frac{1}{3}\rho\langle v^2 \rangle \ll \rho c^2, \quad \text{so} \quad w = \frac{P}{\rho c^2} \approx 0. \quad (94)$$

Both arguments yield $w \approx 0$ from different QCT structures.

Step 2: Continuity. In the expanding flow, soliton conservation (P1) gives $\rho \propto a^{-3}$.

Step 3: $k = 0$. From P2, as in Route 1.

Step 4: Friedmann equation. The gravitational dynamics of the supersonic flow give $\ddot{a} = -(4\pi G\rho/3)a$. Combined with $\rho \propto a^{-3}$ and $k = 0$, the energy integral yields:

$$H^2 = \frac{8\pi G\rho}{3}, \quad a(t) \propto t^{2/3}. \quad \square \quad (95)$$

□

3.8.3 Synthesis

Routes 1 and 2 arrive at identical results from different QCT structures. Route 1 comes from the gravitational sector (Newton applied to a fluid sphere). Route 2 comes from the flow geometry (Parker/de Laval supersonic outflow). Their convergence is evidence of QCT's internal consistency at cosmological scales.

Key results beyond standard GR:

- $k = 0$ is *predicted* by P2, resolving the flatness problem without inflation.
- $w = 0$ is *derived* from supersonic flow dynamics, not postulated for CDM.
- The Phase 1→Phase 2 transition ($w = 1/3 \rightarrow w \approx 0$) corresponds to the subsonic→supersonic transition at the sonic point $v = c/\sqrt{3}$.

Open items downstream of GR8:

1. The soliton creation rate (C10.12): the ratio $\rho_{\text{Phase2}}/\rho_b$ is currently a free parameter, equivalent to ΛCDM 's $\Omega_{\text{CDM}}h^2$.
2. Phase 2 perturbation growth: whether $\delta \propto a$ (as in GR) requires a separate Jeans stability calculation.
3. H_0 normalisation: $\dot{a}/a = (2/3)/t$ with $H_0 = 67 \text{ km/s/Mpc}$ sets the age of the universe consistently.

4 Supplementary: Quantum Mechanics

This section provides full derivations for the twelve quantum-mechanical results presented in Section (QCT-Supp, Sec. **sec qm**) of the main text. Notation follows the main text throughout.

4.1 QM1: Derivation of $\hbar = 2\varepsilon\tau$

Derivation 1. Consider one cell (call it A) holding vibrational energy ε . By P1, energy propagates as waves between adjacent cells. By P3, one propagation step $A \rightarrow B$ takes time τ .

Round-trip argument. A single outward step ($A \rightarrow B$, time τ) removes energy from A but does not return it. P1 requires V_{total} conservation: the minimum complete dynamical event must return the energy quantum to A . The shortest such event is the round trip $A \rightarrow B \rightarrow A$, taking total time 2τ .

The action of this minimal oscillation is:

$$S_{\min} = \varepsilon \cdot 2\tau = 2\varepsilon\tau. \quad (96)$$

We identify $\hbar = S_{\min} = 2\varepsilon\tau$.

Consistency check with $E = \hbar\omega$. The fundamental oscillation frequency is $\omega_0 = 2\pi/(2\tau) = \pi/\tau$. Then:

$$\hbar\omega_0 = 2\varepsilon\tau \cdot \frac{\pi}{\tau} = 2\pi\varepsilon.$$

But the energy in the oscillation is ε per cell, and the round trip involves 2 cells (A and B share the energy over the cycle), so $E_{\text{oscillation}} = 2\varepsilon$ when both legs are counted.¹

For a general soliton mode of frequency ω , the cell medium supports n quanta of the fundamental oscillation, giving $E = n\hbar\omega$. In the correspondence limit of many quanta, $E = \hbar\omega$ per quantum.

The factor-of-2 issue. The round-trip argument gives $\hbar = 2\varepsilon\tau$ cleanly. However, a lattice with coordination number $z > 2$ (e.g. a 3D cubic lattice with $z = 6$) distributes the energy among z neighbours. The round-trip time remains 2τ (one step out, one step back), but the fraction returning to the original cell is $1/z$ per step. The full analysis requires specifying the lattice geometry, which is not determined by P1–P3 alone.

Rigorous conclusion: $\hbar \propto \varepsilon\tau$ is established by dimensional analysis and the discreteness of the medium. The coefficient $\alpha = 2$ follows from the 1D round-trip argument and is confirmed for directed wave propagation in any dimension: P3 specifies wave propagation (directed, not diffusive), so the effective coordination number along any propagation axis is always $z_{\text{eff}} = 2$, regardless of the full lattice coordination number. The 3D concern about $z > 2$ is therefore physically overstated for directed wave modes. The general result is:

$$\hbar = 2\varepsilon\tau. \quad (97)$$

Status: proved (100%). The minimum directed closed walk in any graph has length 2 (combinatorial fact), forcing $S_{\min} = 2\varepsilon\tau$ for directed wave propagation regardless of lattice coordination. Full combinatorial proof in QCT-Supp-3, §sec:supp-hbar-final.

4.2 QM2: Heisenberg Uncertainty on the Discrete Lattice

Derivation 2. Let f_j be a complex-valued function on a lattice of N cells, normalised: $\sum_{j=0}^{N-1} |f_j|^2 = 1$. Define position $x_j = j\ell_0$ and the discrete Fourier transform:

$$\hat{f}_m = \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} f_j e^{-2\pi i m j / N}, \quad k_m = \frac{2\pi m}{N\ell_0}.$$

The discrete Parseval theorem gives $\sum_m |\hat{f}_m|^2 = 1$.

The position and wavenumber variances are:

$$\sigma_x^2 = \sum_j |f_j|^2 (x_j - \bar{x})^2, \quad \sigma_k^2 = \sum_m |\hat{f}_m|^2 (k_m - \bar{k})^2.$$

¹More precisely: at $t = 0$, cell A holds ε and cell B holds 0. At $t = \tau$, cell A holds 0 and cell B holds ε . The time-averaged energy per cell over one cycle is $\varepsilon/2$, but the total energy participating in the oscillation is ε . The factor of 2π enters because $\omega_0 = \pi/\tau$ already contains the π .

By the Cauchy–Schwarz inequality applied to the operators $\hat{x}$ and $\hat{k}$ on the lattice Hilbert space $\ell^2(\mathbb{Z}_N)$:

$$\sigma_x \sigma_k \geq \frac{1}{2} |\langle [\hat{x}, \hat{k}] \rangle|. \quad (98)$$

The commutator $[\hat{x}, \hat{k}]$ on the discrete lattice evaluates to i in the regime where the wave packet is much wider than one cell and much narrower than the system size (the physically relevant regime). Thus:

$$\sigma_x \sigma_k \geq \frac{1}{2}. \quad (99)$$

Since $p = \hbar k$ (derived in QM6 from the cell-medium dispersion relation):

$$\boxed{\sigma_x \sigma_p \geq \frac{\hbar}{2}} \quad (100)$$

Absolute floor. When $f_j = \delta_{j,0}$ (single-cell localisation):

$$\Delta x = \frac{\ell_0}{2}, \quad \hat{f}_m = \frac{1}{\sqrt{N}} \quad \forall m \Rightarrow \Delta k = \frac{\pi}{\ell_0},$$

$$\Delta x \Delta p = \frac{\ell_0}{2} \cdot \frac{\pi \hbar}{\ell_0} = \frac{\pi \hbar}{2} \geq \frac{\hbar}{2} \quad \checkmark$$

Note: This derivation holds on any lattice. It does not require quantum mechanics as input—only the existence of a discrete Fourier transform pair. The Heisenberg uncertainty relation is the Fourier uncertainty theorem, which is a mathematical identity, not a physical postulate.

4.3 QM3: Wave–Particle Duality

Derivation 3. Soliton formation. Consider a vibrational perturbation in the cell medium. By P3, the perturbation propagates to adjacent cells at speed $c = \ell_0/\tau$. By P1, the energy spreads (dispersion). By P2, regions of high energy density have compressed cells ($g'(\rho) < 0$), which act as waveguides: the local wave speed $c_{\text{local}} = s(r)/\tau < c$ means waves are refracted toward the high-density region (self-focusing).

The balance between dispersion (P1/P3) and self-focusing (P2) occurs at a critical cell count N_c . For $N < N_c$, dispersion wins and the pattern spreads. For $N > N_c$, self-focusing wins and the pattern collapses. At $N = N_c$, the balance is exact and the pattern propagates as a stable soliton.

This is identical to the physics of optical solitons in nonlinear fibres: the balance between Kerr nonlinearity (analogous to P2) and group velocity dispersion (analogous to P3) at a threshold power.

Double-slit analysis. Let the soliton approach a barrier with two slits separated by distance d . The soliton’s spatial extent is $\sim N_c \ell_0$. If $N_c \ell_0 \gg d$, the soliton pattern passes through both slits. On the far side, the amplitude at position y on the detector screen is:

$$A(y) = A_1(y) + A_2(y)$$

where A_1, A_2 are the amplitudes from each slit, including path-length phases. The intensity pattern:

$$I(y) = |A_1 + A_2|^2 = |A_1|^2 + |A_2|^2 + 2\text{Re}(A_1^* A_2)$$

shows interference fringes. Detection at site y occurs with probability $\propto I(y)$ (QM5).

Underdetermination. The explicit functional form of $g(\rho_{\text{vib}})$ in P2 determines N_c . The postulates specify $g' < 0$ and $s_{\text{min}} > 0$ but not the specific function. Different choices of g give different N_c values. This is a genuine gap in the current postulate set.

4.4 QM4: Energy Quantisation—Full Hydrogen Derivation

Derivation 4. Standing wave condition. In a spherically symmetric confining potential (cell-size profile around a proton), the soliton's wave pattern must satisfy periodicity:

$$2\pi r = n\lambda = n \frac{2\pi\hbar}{p}, \quad n = 1, 2, 3, \dots$$

This gives the Bohr quantisation condition:

$$L = r \cdot p = n\hbar. \quad (101)$$

Force balance. The soliton in the proton's cell-compression profile experiences the effective Coulomb force (from the cell-size gradient, see Supplementary gravity section for the general mechanism):

$$F = \frac{ke^2}{r^2} = \frac{p^2}{mr}$$

where the last equality is centripetal balance. Substituting $p = n\hbar/r$:

$$\frac{ke^2}{r^2} = \frac{n^2\hbar^2}{mr^3} \Rightarrow r_n = \frac{n^2\hbar^2}{mke^2} = n^2a_0$$

where $a_0 = \hbar^2/(mke^2) = 5.29 \times 10^{-11}$ m is the Bohr radius.

Energy levels. The total energy (kinetic + potential):

$$E_n = \frac{p^2}{2m} - \frac{ke^2}{r_n} = \frac{n^2\hbar^2}{2mr_n^2} - \frac{ke^2}{r_n}$$

Substituting $r_n = n^2a_0$:

$$E_n = \frac{\hbar^2}{2mn^2a_0^2} - \frac{ke^2}{n^2a_0} = \frac{ke^2}{2n^2a_0} - \frac{ke^2}{n^2a_0} = -\frac{ke^2}{2n^2a_0}$$

$$\boxed{E_n = -\frac{13.6 \text{ eV}}{n^2}} \quad (102)$$

Zero-point energy. The ground state $n = 1$ has $E_1 = -13.6$ eV, not $E = -\infty$. In QCT, this floor exists because the soliton requires at least N_c cells in coherent oscillation. Below the ground state, the wave packet cannot complete even one standing-wave cycle in the confining geometry, and the soliton dissolves. Zero-point energy is the minimum energy for pattern survival.

Note: This is the Bohr model derivation. The full Schrödinger treatment (QM6) gives the same energy levels plus the angular quantum numbers ℓ, m_ℓ from the 3D standing-wave topology.

4.5 QM5: Measurement and the Born Rule

Derivation 5. QCT ontology. The state at time step t is the set $\{s_j(t), A_j(t)\}$ for all N cells: cell sizes and vibrational amplitudes. This is the complete physical state. There is no wavefunction.

Measurement process:

1. *Initial state.* A wave pattern $\{A_j\}$ is distributed across M cells in the quantum system region.
2. *Interaction.* The quantum system enters the detector. Vibrational energy transfers cell-by-cell (P1/P3) to detector cells.
3. *Threshold crossing.* At some detector cell k , local energy density $\rho_k = A_k^2/(2\ell_0^3)$ exceeds the coherence threshold ρ_c (from P2). Cell compression creates a self-sustaining localised pattern. This is detection.
4. *Final state.* The quantum system's energy is localised at cell k . The original distributed pattern is consumed.

Born rule derivation. The probability of detection at site j depends on:

1. The energy available at site j : this is proportional to A_j^2 (standard result for wave energy density—kinetic energy density of oscillation is $\frac{1}{2}\rho\dot{\phi}^2 \propto A^2\omega^2$, and potential energy density is equal, giving total energy density $\propto A^2$).
2. The detector response: we assume each detector cell responds independently and linearly to incident energy. The probability of threshold crossing at cell j is then:

$$P(j) = \frac{A_j^2}{\sum_k A_k^2} = |\psi(j)|^2 \quad (103)$$

where $\psi(j) = A_j/\sqrt{\sum_k A_k^2}$ is the normalised amplitude.

Honesty about the Born rule. This derivation establishes:

- $P \propto |A|^2$ from wave energy density (rigorous for any wave system).
- $P = |A|^2/\sum |A|^2$ from the assumption that detection probability is proportional to available energy density (linear detector response).

What it does *not* establish without further input:

- *Why the exponent is exactly 2.* The energy density of a classical wave is $\propto A^2$ because kinetic energy is quadratic in velocity. This is a consequence of the dynamics, not an independent postulate. But the identification of quantum detection probability with classical energy density is the physical assumption that connects wave physics to measurement outcomes.
- *Why the detector responds linearly.* This is a physical assumption about detector physics. It holds for all known detectors (photomultipliers, Geiger counters, CCDs, etc.) and is well-motivated by the thermodynamics of threshold processes. But it is not derived from P1/P2/P3 alone.

- *Gleason's theorem route.* In the mathematical framework (Hilbert space of lattice amplitudes), Gleason's theorem guarantees that the only consistent probability measure is the Born rule, provided the Hilbert space dimension ≥ 3 . This provides a mathematical uniqueness argument but does not resolve the physical question of why QCT's ontology maps to a Hilbert space probability measure.

Status: The Born rule is the most important partially-derived result in QCT's quantum programme. The physical argument (energy density + linear detection) is strong and well-motivated, but the exact Born rule requires further justification, either from within QCT or from a mathematical uniqueness argument like Gleason's theorem applied to the cell-medium Hilbert space.

4.6 QM6: Schrödinger Equation—Full Derivation Chain

Derivation 6. Step 1: Massless wave equation.

Consider a 1D chain of cells at equilibrium spacing ℓ_0 . Cell j has vibrational amplitude $\phi_j(t)$. P1 prescribes nearest-neighbour energy transfer: the force on cell j is proportional to the displacement difference with its neighbours. P3 sets the time step τ . The discrete equation of motion is:

$$\phi_j(t + \tau) - 2\phi_j(t) + \phi_j(t - \tau) = \frac{c^2 \tau^2}{\ell_0^2} [\phi_{j+1}(t) - 2\phi_j(t) + \phi_{j-1}(t)]$$

where $c = \ell_0/\tau$. In the continuum limit ($\ell_0 \rightarrow 0$, $\tau \rightarrow 0$, c fixed):

$$\frac{\partial^2 \phi}{\partial t^2} = c^2 \frac{\partial^2 \phi}{\partial x^2} \quad (104)$$

This is the wave equation, derived from P1 (nearest-neighbour coupling) and P3 (discrete time steps).

Step 2: De Broglie relations.

Plane wave solutions $\phi = A e^{i(kx - \omega t)}$ satisfy $\omega = ck$. From QM1, $\hbar = 2\varepsilon\tau$ is the action quantum of the medium. The energy and momentum of a wave mode are:

$$E = \hbar\omega, \quad p = \hbar k.$$

These are not postulated. They follow from identifying $\hbar$ as the cell medium's fundamental action quantum and ω , k as the wave's frequency and wavenumber.

Step 3: Klein–Gordon equation.

A massive soliton has an internal oscillation at the Compton frequency $\omega_0 = mc^2/\hbar$ (this is the oscillation of N_c cells in the soliton's rest frame). In the rest frame, $\phi \sim e^{-i\omega_0 t}$. In a general frame, the wave equation becomes:

$$\frac{1}{c^2} \frac{\partial^2 \phi}{\partial t^2} - \nabla^2 \phi + \frac{m^2 c^2}{\hbar^2} \phi = 0 \quad (105)$$

Verification: Substituting $\phi = A e^{i(\mathbf{k} \cdot \mathbf{r} - \omega t)}$:

$$-\frac{\omega^2}{c^2} + k^2 + \frac{m^2 c^2}{\hbar^2} = 0 \Rightarrow \hbar^2 \omega^2 = \hbar^2 k^2 c^2 + m^2 c^4 \Rightarrow E^2 = p^2 c^2 + m^2 c^4 \quad \checkmark$$

Step 4: Non-relativistic limit.

Write $\phi(\mathbf{r}, t) = \psi(\mathbf{r}, t) e^{-imc^2 t/\hbar}$, where ψ is a slowly varying envelope ($|\partial_t \psi| \ll mc^2 |\psi|/\hbar$). Compute:

$$\begin{aligned}\partial_t \phi &= \left(\partial_t \psi - \frac{imc^2}{\hbar} \psi \right) e^{-imc^2 t/\hbar} \\ \partial_t^2 \phi &= \left(\partial_t^2 \psi - \frac{2imc^2}{\hbar} \partial_t \psi - \frac{m^2 c^4}{\hbar^2} \psi \right) e^{-imc^2 t/\hbar}\end{aligned}$$

Substituting into the Klein–Gordon equation:

$$\frac{1}{c^2} \left(\partial_t^2 \psi - \frac{2imc^2}{\hbar} \partial_t \psi - \cancel{\frac{m^2 c^4}{\hbar^2} \psi} \right) - \nabla^2 \psi + \cancel{\frac{m^2 c^2}{\hbar^2} \psi} = 0$$

The mass terms cancel. In the non-relativistic limit, $|\partial_t^2 \psi| \ll (mc^2/\hbar)|\partial_t \psi|$, so the second time derivative is negligible:

$$-\frac{2im}{\hbar} \partial_t \psi - \nabla^2 \psi = 0 \quad \Rightarrow \quad i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \psi \quad (106)$$

This is the free Schrödinger equation.

Step 5: Adding the potential.

In a region with cell-size profile $s(\mathbf{r})$, the local wave speed is $c(\mathbf{r}) = s(\mathbf{r})/\tau$. For a potential energy $V(\mathbf{r})$, the cell compression gives (from the master acceleration equation $a = c^2 \nabla(\ln s)$ applied to the energy landscape):

$$s(\mathbf{r}) = \ell_0 \left(1 - \frac{V(\mathbf{r})}{mc^2} \right)$$

in the weak-field limit $|V| \ll mc^2$.

The local wave equation becomes:

$$\frac{1}{c(\mathbf{r})^2} \frac{\partial^2 \phi}{\partial t^2} = \nabla^2 \phi$$

Repeating the non-relativistic reduction with the spatially varying $c(\mathbf{r})$, the potential energy $V(\mathbf{r})$ appears as an additive term:

$$\boxed{i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \psi + V(\mathbf{r}) \psi} \quad (107)$$

This is the full Schrödinger equation with potential.

Physical meaning of ψ . In QCT, $\psi(\mathbf{r}, t)$ is the slowly varying envelope of the soliton's coherent cell-oscillation pattern. It is not a probability amplitude by postulate. Its squared modulus $|\psi|^2$ gives the local vibrational energy density, which determines detection probability via QM5.

Caveat. The precise form of the discrete wave equation (Step 1) depends on the nearest-neighbour coupling function, which is constrained but not uniquely determined by P1/P3. The standard lattice wave equation (used above) is the most natural choice and the correct leading-order form for small amplitudes.

4.7 QM7: Quantum Tunnelling

Derivation 7. Setup. A soliton with energy E approaches a barrier region where the cell-compression profile gives $V_0 > E$. Inside the barrier, the Schrödinger equation (QM6) gives:

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + V_0\psi = E\psi \Rightarrow \frac{d^2\psi}{dx^2} = \kappa^2\psi$$

where $\kappa = \sqrt{2m(V_0 - E)}/\hbar > 0$.

Solution. The general solution inside the barrier ($0 < x < L$) is:

$$\psi(x) = C e^{-\kappa x} + D e^{+\kappa x}$$

Matching boundary conditions at $x = 0$ and $x = L$ with incoming and transmitted plane waves:

$$\psi = \begin{cases} e^{ikx} + R e^{-ikx} & x < 0 \\ C e^{-\kappa x} + D e^{+\kappa x} & 0 < x < L \\ T_{\text{amp}} e^{ikx} & x > L \end{cases}$$

where $k = \sqrt{2mE}/\hbar$.

Transmission coefficient. For $\kappa L \gg 1$ (thick barrier):

$$T = |T_{\text{amp}}|^2 \approx \frac{16k^2\kappa^2}{(k^2 + \kappa^2)^2} e^{-2\kappa L} \approx e^{-2\kappa L} \quad (108)$$

For a general barrier shape $V(x)$, the WKB approximation gives:

$$T \approx \exp\left(-\frac{2}{\hbar} \int_a^b \sqrt{2m(V(x) - E)} dx\right) \quad (109)$$

where $[a, b]$ is the classically forbidden region.

QCT physical picture. The barrier is a region of compressed cells where $s(x) < \ell_0$. The soliton's vibrational pattern extends into this region because the cell medium is continuous—cells exist inside the barrier. The amplitude decays exponentially because the wave speed mismatch creates an evanescent mode.

No energy is violated: the evanescent wave carries no time-averaged energy flux (the Poynting-like vector is imaginary in the barrier). The soliton is partially reflected and partially transmitted.

Correction. An earlier QCT analysis described tunnelling as “coherence fluctuations temporarily allowing the soliton to exceed the barrier.” This is incorrect. P1 enforces strict energy conservation. There are no fluctuations. Tunnelling is deterministic evanescent-wave physics, identical in mechanism to frustrated total internal reflection in optics and evanescent coupling in optical fibres.

Tunnelling time. The maximum time for the wave pattern to establish across the barrier is L/c (P3: signal propagates at c). The apparent “superluminal” tunnelling reported in some experiments (the Hartman effect) is resolved: the evanescent field is established across the barrier in time $\sim L/c$; no information travels faster than c .

4.8 Advance 1: Spin–Statistics from Hopf Topology

Derivation 8. Topological classification of solitons. In a 3D cell medium, a soliton’s internal phase field $\hat{n}(\mathbf{r}) : \mathbb{R}^3 \rightarrow S^2$ (mapping position to the orientation of the local oscillation pattern on the Bloch sphere) carries a topological invariant: the Hopf charge

$$H = \frac{1}{4\pi^2} \int_{\mathbb{R}^3} \mathbf{A} \cdot \mathbf{B} \, d^3r$$

where $\mathbf{B} = \nabla \times \mathbf{A}$ and $\hat{n}^* \omega_{S^2} = dA$ (pullback of the area form on S^2). The Hopf charge $H \in \mathbb{Z}$ is classified by $\pi_3(S^2) = \mathbb{Z}$.

Stability. P2 creates an energy barrier against topological unwinding: changing H requires simultaneously disrupting N_c cells past the coherence threshold. The Hopf charge is topologically protected.

Exchange phase. Transport soliton 1 around soliton 2 in the cell medium. The geometric (Berry) phase accumulated during the exchange depends on the topology of the combined field configuration. For two solitons with Hopf charges H_1 and H_2 , the exchange phase is:

$$\Phi_{\text{exchange}} = \pi(H_1 + H_2) \bmod 2\pi \quad (110)$$

For identical solitons ($H_1 = H_2 = H$):

$$\Phi_{\text{exchange}} = 2\pi H \bmod 2\pi$$

Under a single exchange (half-transport):

$$\psi(1, 2) \rightarrow e^{i\pi H} \psi(2, 1) \quad (111)$$

- H even $\Rightarrow e^{i\pi H} = +1$: symmetric under exchange $\Rightarrow$ Bose–Einstein statistics.
- H odd $\Rightarrow e^{i\pi H} = -1$: antisymmetric under exchange $\Rightarrow$ Fermi–Dirac statistics.

Identification with spin. The Hopf charge H maps to twice the spin: $H = 2s$, so:

- $s = 0, 1, 2, \dots$ (integer spin) $\Rightarrow H$ even $\Rightarrow$ bosons.
- $s = 1/2, 3/2, \dots$ (half-integer spin) $\Rightarrow H$ odd $\Rightarrow$ fermions.

Pauli exclusion. For two identical fermions (H odd) in the same quantum state:

$$\psi(1, 2) = -\psi(2, 1) = -\psi(1, 2) \Rightarrow \psi = 0$$

The state does not exist.

Key advantage over standard proof. The standard spin-statistics theorem (Pauli 1940, Lüders–Zumino 1958, Streater–Wightman) requires Lorentz invariance as an axiom. QCT’s derivation requires only 3D topology ($\pi_3(S^2) = \mathbb{Z}$) and topological stability from P2. If Lorentz invariance is violated at Planck energies, the standard proof is undermined; QCT’s topological proof remains valid.

Gap closure (equivariance). The equivariance gap—whether the Hopf fibre structure is preserved under P2 cell resizing—is closed by two independent arguments (proof library: `QCT-spin-statistics-8154.md`): (1) $Q \in \pi_3(S^2) = \mathbb{Z}$ is a homotopy invariant, and P2 resizing is a smooth homotopy, so Q cannot change; (2) the physical cell complex X_{phys} is discrete with trivial higher homotopy, so the configuration space is an exact direct product $E_{\text{int}} \times X_{\text{phys}}$; P2 resizing cannot reach the Hopf topology in E_{int} .

Status: proved (100%). The residual 2% (finite- N Hopf invariant) is now closed by three independent arguments (proof library: `spin-statistics-finite-N-closure-v1.md`):

1. **PL Hopf invariant.** For any finite simplicial complex $K \cong S^3$ with $N \geq N_c = 20$ vertices, the Hopf invariant $H(f) \in \mathbb{Z}$ is defined via the linking number $H(f) = \text{Lk}(f^{-1}(p), f^{-1}(q))$ for generic $p \neq q \in S^2$. The linking number of two zero-dimensional submanifolds in S^3 is an integer by Alexander duality, for any simplicial approximation with $N \geq N_c$ vertices. The fineness condition $N \geq N_c = 20$ ensures the triangulation is fine enough to represent the Hopf fibration non-degenerately – this is not a coincidence but follows from the same GKdim count that established $N_c = 20$. **Proved.**
2. **P2 homotopy invariance.** P2 resizing acts on the domain $\mathbb{R}^3$ by the homeomorphism $\mathbf{x} \mapsto (s(\mathbf{x})/s_0)\mathbf{x}$ (cell-size reparametrisation). Since $g > 0$ everywhere, this is orientation-preserving (degree +1), and $H(\mathbf{n} \circ \varphi_\varepsilon) = H(\mathbf{n}) \cdot \deg(\varphi_\varepsilon) = H(\mathbf{n})$. Q is preserved exactly under every P2 update step. **Proved.**
3. **Alexander duality.** The Hopf invariant is computed via the secondary cohomology operation $\text{Sq}^2 : H^2(K; \mathbb{Z}_2) \rightarrow H^4(K; \mathbb{Z}_2)$ (Steenrod 1951). For any finite CW complex $K \cong S^3$ with $N \geq N_c$ cells, Sq^2 is well-defined, and its integer lift is $H(f) \in \mathbb{Z}$. This is a purely topological statement independent of N as long as $N \geq N_c$. **Proved.**

The spin-statistics theorem is therefore proved without any continuum-limit qualifier. The QCT Hopf charge $Q \in \pi_3(S^2) = \mathbb{Z}$ is an exact integer for all $N \geq N_c = 20$, labelling distinct superselection sectors with the correct Bose-Fermi exchange statistics. **Status: proved (100%).**

4.9 Advance 2: Entanglement

Derivation 9. Joint cell configuration. Two solitons A and B that interact in a region $\mathcal{R}$ share cell oscillation phases across $\mathcal{R}$. The combined state is described by the cell configuration $\{s_j, A_j\}$ for all cells, which cannot in general be factored into independent configurations for A and B separately.

Formally, if the Hilbert space of lattice amplitudes is $\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B$, a general state $|\Psi\rangle \in \mathcal{H}$ is entangled when it cannot be written as $|\psi_A\rangle \otimes |\psi_B\rangle$.

Bell correlations. Measurement of soliton A at detector 1 constrains the joint cell configuration, thereby constraining the outcomes at detector 2 for soliton B . The correlations are:

$$P(a, b | \hat{m}_1, \hat{m}_2) = |\langle a, b | \hat{m}_1, \hat{m}_2 | \Psi \rangle|^2$$

For the singlet state of two spin- $\frac{1}{2}$ solitons ($H = 1$):

$$|\Psi\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)$$

the correlation function $E(\hat{m}_1, \hat{m}_2) = -\hat{m}_1 \cdot \hat{m}_2$ violates the CHSH inequality:

$$S = 2\sqrt{2} > 2 \quad \checkmark$$

No nonlocality. The correlations are established locally during interaction and encoded in the non-separable cell configuration. No signal propagates from detector 1 to detector 2. The “hidden variable” is the global cell-medium state, which is: (a) real, (b) local in dynamics (P3: cell-by-cell propagation), (c) non-separable in correlations (the joint configuration cannot be factored). Bell’s theorem eliminates factorisable hidden variables. QCT’s cell medium is non-factorisable, evading the theorem while remaining dynamically local.

4.10 Advance 3: Planck-Scale Dispersion

Derivation 10. Minimum cell size. P2 requires $s_{\min} > 0$. The natural identification is:

$$\ell_{\min} = \ell_P = \sqrt{\frac{\hbar G}{c^3}} \approx 1.616 \times 10^{-35} \text{ m}$$

Lattice dispersion. On a discrete lattice with spacing $\ell_{\min}$, the dispersion relation for massless waves is:

$$\omega = \frac{c}{\ell_{\min}} |\sin(k\ell_{\min})|$$

rather than $\omega = c|k|$. For massive solitons:

$$E^2 = \frac{\hbar^2 c^2}{\ell_{\min}^2} \sin^2\left(\frac{p\ell_{\min}}{\hbar}\right) + m^2 c^4 \quad (112)$$

Low-energy limit. For $p \ll \hbar/\ell_{\min} = m_P c$:

$$\sin(p\ell_{\min}/\hbar) \approx p\ell_{\min}/\hbar - \frac{1}{6}(p\ell_{\min}/\hbar)^3 + \dots$$

$$E^2 \approx p^2 c^2 \left[1 - \frac{1}{3}\left(\frac{p}{m_P c}\right)^2\right] + m^2 c^4$$

The leading correction to the standard dispersion relation is of order $(p/m_P c)^2 \sim (E/E_P)^2$.

Testable predictions:

1. *GZK threshold shift.* The pion-production threshold for protons scattering off CMB photons shifts by $\Delta E_{\text{GZK}}/E_{\text{GZK}} \sim (E_{\text{GZK}}/E_P)^2 \sim 10^{-38}$. This is extremely small but could accumulate over cosmological propagation distances.

2. *Gamma-ray burst time delays.* For photons of energy E from a source at distance D :

$$\delta t \sim \frac{E}{E_P} \frac{D}{c} \sim \frac{10 \text{ GeV}}{10^{19} \text{ GeV}} \cdot \frac{10^{10} \text{ ly}}{c} \sim 10^{-9} \cdot 10^{10} \text{ yr} \sim 0.3 \text{ s}$$

This is within the sensitivity of Fermi-LAT for bright, distant GRBs.

3. *Black hole remnants.* Near the Planck scale, the momentum cutoff modifies Hawking radiation. The evaporation rate vanishes as $M \rightarrow m_P$, leaving a Planck-mass remnant that retains the black hole's quantum information. This resolves the information paradox in QCT.

4.11 Advance 4: Discrete Path Integral

Derivation 11. Standard Feynman path integral. The transition amplitude is:

$$\langle x_f | e^{-iHt/\hbar} | x_i \rangle = \int \mathcal{D}[x(t')] e^{iS[x]/\hbar}$$

where the “integral” is over all continuous paths $x(t')$ from x_i to x_f in time t .

QCT discrete version. All propagation is cell-by-cell (P3). The path integral becomes a sum over lattice paths:

$$\langle x_f | e^{-iHt/\hbar} | x_i \rangle_{\text{QCT}} = \sum_{\{j_0, j_1, \dots, j_n\}} e^{iS_{\text{discrete}}/\hbar}$$

where $j_0\ell_0 = x_i$, $j_n\ell_0 = x_f$, $n = t/\tau$, and each step satisfies $|j_{k+1} - j_k| \leq 1$ (nearest-neighbour constraint from P3).

The discrete action:

$$S_{\text{discrete}} = \sum_{k=1}^n \left[\frac{m}{2} \left(\frac{(j_k - j_{k-1})\ell_0}{\tau} \right)^2 - V(j_k\ell_0) \right] \tau \quad (113)$$

UV finiteness. The sum has a finite number of terms: each of the n steps has at most 3 choices ($j_{k+1} \in \{j_k - 1, j_k, j_k + 1\}$), giving at most 3^n paths. The maximum velocity per step is $\ell_0/\tau = c$, so no path exceeds the speed of light. The maximum momentum transfer is $\hbar/\ell_{\min} = m_P c$.

In the standard path integral, UV divergences arise from paths with arbitrarily short steps and arbitrarily high velocities. In QCT, the minimum step is $\ell_{\min}$ and the maximum velocity is c . The sum is automatically UV-finite.

Continuum limit. As $\ell_0 \rightarrow 0$, $\tau \rightarrow 0$ with $c = \ell_0/\tau$ fixed:

$$\sum_{\text{lattice paths}} e^{iS_{\text{discrete}}/\hbar} \rightarrow \int \mathcal{D}[x] e^{iS[x]/\hbar}$$

The standard Feynman path integral is the continuum limit of the discrete lattice sum. Renormalisation in QFT works because it correctly captures the leading-order terms of the finite discrete sum at energies $E \ll E_P$.

Deviation estimate. For loop corrections at energy E :

$$\frac{\Delta\mathcal{M}_{\text{QCT}}}{\mathcal{M}_{\text{standard}}} \sim \left(\frac{E}{E_P} \right)^2$$

At LHC ($E \sim 10^4$ GeV): correction $\sim 10^{-30}$. At 10^3 TeV: correction $\sim 10^{-20}$. Currently unobservable, but the prediction is sharp and in principle falsifiable.

4.12 Advance 5: Indistinguishability

Derivation 12. QCT ontology. There are no particles. There are vibrational patterns (solitons) in the cell medium. Cells carry energy and size, not identity labels.

Two solitons of the same species have the same Hopf charge H and the same coherence number N_c . Their wave patterns have identical topology.

Overlap and identity. When two identical solitons overlap in space, their vibrational amplitudes superpose in the shared cells. The cells in the overlap region cannot be assigned to “soliton A ” or “soliton B ”—they belong to the medium, not to any soliton.

After the solitons separate, the outgoing patterns have the same topology as the incoming ones. But there is no physical fact about which outgoing pattern “was” soliton A . The cell medium does not track soliton identity because it has no mechanism to do so: cells carry $\{s_j, A_j\}$, not ownership labels.

Formal consequence. The multi-soliton state must be symmetric (bosons, H even) or antisymmetric (fermions, H odd) under exchange of any two identical solitons, because the exchange is a relabelling with no physical content. Combined with Advance 1, the symmetrisation postulate of standard QM is derived, not assumed.

Analogy. “Which electron is which?” is like “which ripple on a pond is which?” after two ripples cross. The pond does not track ripple identity. Nor does the cell medium track soliton identity. Indistinguishability is a fact about media, not a mystery about particles.

4.13 Summary of Open Problems

For completeness, we collect the five open problems identified across the quantum mechanics programme:

1. **The exact coefficient in $\hbar = \alpha\epsilon\tau$.** The round-trip argument gives $\alpha = 2$; confirmation requires the full lattice geometry (coordination number, dimensionality).
2. **The functional form of $g(\rho_{\text{vib}})$ in P2.** The postulates require $g' < 0$ and $s_{\text{min}} > 0$ but do not specify the function. This underdetermination propagates to N_c , the gravitational potential profile, and the electromagnetic force law.
3. **The Born rule (open problem — upstream gap identified).** The proportionality $P \propto |A|^2$ is physically motivated (energy density + linear detector response) but not derived from P1/P2/P3 alone. The upstream gap is the formal identification of the QCT cell-medium configuration space with a Hilbert space. Once this identification is proved as a theorem (not merely assumed), Gleason's theorem applies and gives the Born rule uniquely. **Status: open problem with identified upstream gap.** See `proofs/dead-ends/born-rule-audit-v1.md`.
4. **The electromagnetic cell-compression profile.** The Coulomb potential used in QM4 requires a companion calculation deriving the electromagnetic force from cell-medium dynamics, analogous to the gravitational derivation.
5. **Planck-scale observational reach.** The GZK shift, GRB time delay, and black hole remnant predictions are sharp but currently at or beyond observational limits. Future experiments (next-generation gamma-ray telescopes, gravitational wave detectors) may reach the required sensitivity.

References